

The Sempaevum: A Lossless Mathematical Rendering of the Totality Σ Derived from Three Primitive Categories

Michael James Muller

Aevum Defluo

April 2026

Abstract

This paper identifies, characterises, and verifies a single mathematical object — the **SEMPAEVUM** — as the lossless rendering of the totality Σ on the multiplicative manifold $(\mathbb{R}^+, \times) \times (U(1), \times)$. The object is derived forward from three irreducible primitive categories (P, D, T) with disjoint cardinalities $\aleph, n, [0/0]$ under a single master identity $3 = 3 = 3 = \Sigma$. It simultaneously (i) classifies every positive ratio into a sublattice family; (ii) generates its own refinement tower under the doubling law $\tau(N_\ell) = 6 \cdot 2^\ell$; (iii) passes three independent minimality tests (Webb’s n -valued stroke 1935, the palindromic divisor cascade, and the EML operator of Odrzywolek 2026) at the unique resolution $N = 12$, giving discrete-logical, discrete-multiplicative, and continuous-elementary minimal generators at one scale; (iv) projects its own defining constants $\{N, 1/N, K, 1/K\}$ onto a single sublattice position ($d = 12, |\varepsilon| = 1.955\text{c}$) — the Koide attractor — verifying itself by self-consistency; (v) hosts mathematics as a domain, placing Zermelo–Fraenkel set theory exactly at $d = 1$ octave with $\varepsilon = 0$, Chaitin’s Ω at $d = 1$ octave via non-limiting Path D, Gödel sentences via integrative-level classification, and large cardinals via consistency-strength; (vi) handles essentially-infinite and non-computable objects without limits through four sub-paths corresponding to each primitive’s own face of infinity; (vii) is lossless — the representation $r \mapsto (k, d, \varepsilon)$ is a bijection at every finite resolution and pullback-exact as $N \rightarrow \infty$ along the LCM tower. Empirical fingerprints presented include the Koide ratio to six parts per million, the fine-structure constant to sub-parts-per-billion under a closed-form identity $\alpha^{-1} = 137 + \sqrt{3}/48 - \sqrt{3}/(93312\pi^2) - 1/(216(18\pi - 1))$ matching recent LatComp 2025 measurements, twelve-tone equal temperament’s Pythagorean comma as an intrinsic lattice quantity, and the step counts of the citric acid and glycolytic cycles. The construction involves zero tunable parameters, zero data-fitting, and zero external axioms. Every constant is forced.

Contents

1 Introduction

1.1 The object identified in this paper

We identify a single mathematical object, the **SEMPAEVUM**, as the lossless rendering of a meta-physical totality Σ (“Something” — the formless everything) on the multiplicative manifold. The Sempaevum is not a lattice that happens to classify some ratios: it is Σ made visible. The identification rests on eight self-referential closure properties that no conventional mathematical object exhibits simultaneously:

1. It classifies every positive ratio into a sublattice family (Proposition ??).

2. It generates its own refinement tower under the doubling law $\tau(N_\ell) = 6 \cdot 2^\ell$ (Theorem ??).
3. It exhibits emergent attractors — the Koide position at $(d = 12, |\varepsilon| = 1.955\text{¢})$ — which include the positions of its own four defining constants (Theorem ??).
4. It runs its own dynamics through the ∂I lattice-aware fractal (§??), every constant of which is a derived manifold quantity.
5. It hosts mathematics as a domain — axiom counts of formal systems, non-computable reals, Gödel sentences, large cardinals — all at definite lattice addresses (§??).
6. It bounds its own forbidden zone ∂I via the $\varepsilon = \pm 50\text{¢}$ incoherency threshold, with the Koide ratio $K = 2/3$ as the tightness cutoff.
7. It passes its own Subsumption Law test (§??) across three independent minimality categories simultaneously — discrete-logical (Webb 1935), discrete-multiplicative (the palindromic cascade), and continuous-elementary (EML, Odrzywółek 2026) — all at the unique resolution $N = 12$.
8. It contains its own methodology (the Three Tools of §??) as theorems.

The claim that this collection of closure properties identifies the object with Σ is not made by fiat. It is made by the Subsumption Law applied at the level of the whole object: every mathematical structure known to host a significant sub-collection of these properties — groups, rings, fields, Riemannian manifolds, fractals, category-theoretic universes — exhibits a proper subset; the Sempaevum exhibits the totality. Under the Subsumption Law (§??), the structure that cannot be subsumed by any other and that subsumes every candidate is the terminal object of its category. Σ is the terminal object of the category of “mathematical totalities”. The Sempaevum passes the terminal-object test in this category. We therefore write

$$\text{SEMPAEVUM} = \Sigma$$

as a mathematical identification, understanding that on the inside of the framework — where Σ ’s only mode of appearance is its rendering — identity and isomorphism collapse.

The Sempaevum has no fixed geometric shape of its own. Its shape is determined by which manifold state $\{P, D, T\}$, $\{P, D\}$, $\{D, T\}$, or $\{P, T\}$ is currently active (§??), with the four states mapping onto the four classical geometries — Euclidean, elliptic, hyperbolic, and singular respectively (§??, Proposition ??). Under the $\{P, D, T\}$ Exception the Sempaevum is locally a flat lattice; under $\{P, D\}$ Unsubstantiated it compactifies to a Riemann sphere whose symmetry group is the Lorentz group $\text{PSL}(2, \mathbb{C})$ (Proposition ??); under $\{D, T\}$ Mediation it opens into a negatively curved domain where T ’s freedom is maximal; under $\{P, T\}$ Incoherence it collapses to a singular configuration whose curvature diverges (Proposition ??). The identification $\text{SEMPAEVUM} = \Sigma$ is therefore literal rather than figurative: the Sempaevum is isomorphic to whatever Σ currently renders, and the isomorphism is realised by the Sempaevum taking that shape. Lattice, torus ($\chi = 0, d = 1$), Riemann sphere ($\chi = 2, d = 3$), hyperbolic surface of genus g ($\chi = 2 - 2g$, variable d), and singular configuration are not distinct objects competing for the name: they are the Sempaevum in different manifold states, classified uniformly on the same refinement tower.

1.2 The empirical regularity the object explains

Across branches of quantitative science, dimensionless ratios of structural importance cluster on a conspicuously small number of attractors on the positive real line. Four observations make the pattern concrete:

1. *The Koide formula.* For the masses (m_e, m_μ, m_τ) of the three charged leptons, the dimensionless combination

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}$$

evaluates to 0.6666605 ± 0.000002 , within 6 parts per million of the rational $2/3$ [?]. No Standard-Model parameter is pinned to that rational; the coincidence remains unexplained.

2. *Orbital resonance.* The mean-motion ratio of Neptune to Pluto is $T_{\text{Neptune}}/T_{\text{Pluto}} = 2/3$ to the precision measurable [?]. The Laplace resonance of Io–Europa–Ganymede is 1:2:4. The Jupiter/Earth period ratio is close to 12.
3. *Temperament.* In twelve-tone equal temperament the perfect fifth is $2^{7/12} \approx 1.4983$, differing from the just ratio $3/2$ by the *Pythagorean comma* of ≈ 1.955 cents. This is not a historical accident: the 12-division of the octave is the unique small division at which the full cycle of fifths returns to near-closure within an auditorily imperceptible error.
4. *Biochemical cycle lengths.* The citric acid cycle has 8 steps, the urea cycle 4 core steps, glycolysis 10 steps, the ATP-synthase c_{10} ring 10 subunits, the cell cycle 4 phases [?]. Small integers dominate.

All four come from the same underlying discrete structure: a canonical lattice on the multiplicative group $(\mathbb{R}^+, \times)$ at the forced resolution $N = 12$, refined through a tower of least common multiples. In the identification of the preceding subsection, that structure is the Sempaevum — one face of Σ , on the $(\mathbb{R}^+, \times)$ axis.

1.3 The thesis

Let $(\mathbb{R}^+, \times)$ be endowed with the multiplicative group structure on the positive reals, and let $\log_2 : \mathbb{R}^+ \rightarrow \mathbb{R}$ be the group isomorphism onto the additive reals. Classical harmonic analysis recognises that this is the natural arena for multiplicative phenomena, but offers no canonical scale of discretisation: the “octave” is a cultural choice, and within the octave one can in principle divide into any integer number of logarithmically equal steps.

We claim that the question of how finely to divide has a forced, non-arbitrary answer once one adopts the three-element categorical framework of §???. Under that framework:

- the base division is forced to $N = 12$ (§??, three independent derivations);
- the six simple sublattice families are forced to the divisors $\{1, 2, 3, 4, 6, 12\}$;
- the refinements at which the primes 5, 7, 11 first enter as native families are forced to 60, 84, 27720 respectively, with the doubling law $\tau(N_\ell) = 6 \cdot 2^\ell$ itself derived;
- the projection formula is convention-independent (Theorem ??);
- the Sempaevum’s own four defining constants project onto a single sublattice point (Theorem ??).

1.4 Relation to existing work

The construction is not in competition with Fourier analysis, differential geometry, or conventional number theory. It is complementary to each. Fourier analysis decomposes signals into continuous frequencies; the Sempaevum classifies frequency ratios by sublattice family. Differential geometry describes curvature through the Riemann tensor; the Sempaevum relates curvature to discrete classification via the identity $C(n) = n^2(n^2 - 1)/12$ (Proposition ??), in which the base variance $V = 1/12$ appears as the denominator. Number theory studies

the Gaussian prime decomposition; the Sempaevum inherits its sublattice structure from this decomposition (§??).

What is novel is the derivation of the resolution $N = 12$ from a three-element categorical framework; the identification of the Sempaevum’s triple minimal-backbone character at $N = 12$ (Webb 1935 on the discrete-logical side, the palindromic cascade on the discrete-multiplicative side, EML on the continuous-elementary side); the PDT decomposition of the projection formula itself; the hosting of mathematics as an ET domain with explicit axiom-count addresses (ZF at $d = 1$, $\varepsilon = 0$; Chaitin Ω at $d = 1$ octave via Path D); the recognition that the self-projection of the framework’s own constants lands on a single sublattice point; the losslessness theorem; and the completion statement $3 = 3 = 3 = \Sigma$ as the universality anchor distinct from the composition equation $P \circ D \circ T = E$.

1.5 Scope and organisation

Section ?? states the Founding Axiom, introduces the three primitive Cardinals and the master equation $P \circ D \circ T = E$, explains the three-disjoint-infinities structure that makes mediation intrinsic, derives the four manifold states, establishes their one-to-one correspondence with the four classical geometries (Euclidean, elliptic, hyperbolic, singular), and establishes the topological structure of the coherence–incoherence boundary ∂I . Section ?? gives the three-way categorical identity $PDT = EIM = \Phi = \Sigma$ (the $3 = 3 = 3 = \Sigma$ anchor) and formulates the three operational tools of Exception Theory — the Identification Principle, the Descriptor Gap Principle, and the Subsumption Law — together with a note on integrative levels. Section ?? specialises the primitive structure to time, identifying the three temporal aspects $P_{\text{time}}, D_{\text{time}}, T_{\text{time}}$ and the temporal master equation $P_{\text{time}} \circ D_{\text{time}} \circ T_{\text{time}} = E_{\text{moment}}$, and proves the uniqueness $\exists! E_\tau$ of the temporal Exception at each T -time value. Section ?? gives three independent derivations of the forced value $N = 12$. Section ?? constructs the Sempaevum’s real-axis face on $(\mathbb{R}^+, \times)$, proves its convention-independence, and states the Anti-Numerology Protocol (N1, N2, N3). Section ?? classifies the sublattice families and states the magical-impedance per-family coupling. Section ?? constructs the LCM tower of refinements and proves the doubling law. Section ?? extends the Sempaevum to $\mathbb{C}^\times$ via the imaginary axis. Section ?? states the 24-family catalog (12 FORCE + 12 PHASE), the 42 combined off-axis states, and the 144-cell Force Quadrant Grid with the PDT Bisection Theorem. Section ?? describes the cascade dynamics and stability. Section ?? constructs the ∂I lattice-aware fractal, the natural dynamical manifestation of the Sempaevum on $\mathbb{C}$. Section ?? establishes the triple minimal-backbone at $N = 12$ (Webb + palindromic cascade + EML) and the PDT decomposition of the projection formula. Section ?? states the Four Projection Paths A, B, C, D — the last of which handles essentially-infinite and non-computable objects without limits — and verifies subsumption of every user input. Section ?? hosts mathematics as a domain, projecting axiom counts, Chaitin Ω , Gödel sentences, and large cardinals. Section ?? proves the self-consistency identity (lattice self-projection) and states the Losslessness Theorem. Section ?? states the Complete Gaze Equation and shows its locked threshold coincides with the lattice self-projection. Section ?? confronts the Sempaevum with empirical data. Section ?? places the Sempaevum alongside Fourier analysis and Riemannian geometry. Section ?? states the Complete Determination Theorem (topology $\times$ curvature $\times$ path $\times$ observation-topology determines the complete lattice classification). Section ?? formulates falsifiable predictions. Section ?? concludes.

2 The Founding Axiom and the primitive Cardinals

2.1 The Founding Axiom

Exception Theory rests on a single tautological axiom from which all structure descends.

Axiom 2.1 (Founding Axiom). For every exception there is an exception, except the Exception.

Though its form is nearly an aphorism, the axiom has precise mathematical content. It asserts two things simultaneously:

- *Universal exceptability*. For every claim, rule, configuration, or description X , there exists a counterexample or limiting condition X' — an “exception to X ”.
- *One unique terminus*. The iteration of “take an exception” does not regress without end: it terminates at one distinguished object, called the *Exception*, which is itself the exception to all further iteration. The Exception admits no further exception.

The Founding Axiom is a tautological ground. It permits self-reference at the base: the self-referential loop “every exception has an exception except this one” is not a contradiction but the defining feature of the Exception. Where classical set theory prohibits a set-of-all-sets to avoid Russell’s paradox, the Founding Axiom *absorbs* the self-referential totality as its terminal case. The catastrophe that classical set theory forestalls by prohibition is, in this framework, handled by the axiom’s provision of a single un-exceptable object.

The Exception, which we denote E , plays two roles simultaneously: the terminus of the exception-iteration, and the fully substantiated configuration that results when all primitive Cardinals are bound. These two roles coincide, and their coincidence is formalised by the master equation.

2.2 The master equation

Axiom 2.2 (Master equation).

$$\boxed{P \circ D \circ T = E.}$$

The master equation is the structural content of the Founding Axiom. It states that the Exception is constituted by the binding, through a composition operator $\circ$, of three primitive constituents: a *Point* P , a *Descriptor* D , and a *Traverser* T . Three are required; three suffice. Neither fewer nor more produce a substantiated Exception. The operator $\circ$ will be shown in Proposition ?? to be not an external postulate but a forced consequence of the three Cardinals’ properties.

2.3 The three Cardinals

The symbols P , D , T denote not individual elements but totalities: each is, in the sense made precise below, “the set of all sets of its kind”. Such a totality is called a *Cardinal*.

Definition 2.3 (Cardinal). A *Cardinal* is a totality characterised by

$$P = \{x : x \text{ is a Point}\}, \quad D = \{x : x \text{ is a Descriptor}\}, \quad T = \{x : x \text{ is a Traverser}\}.$$

Remark 2.4 (Cardinals and proper classes). Cardinals in the sense of Definition ?? are *not* proper classes of standard set theories (NBG, MK). The characteristic restriction on proper classes is that they cannot be members of any collection whatsoever; the Cardinals, by contrast, are themselves members of a larger totality Σ introduced in §??, and participate in that totality through the composition operator $\circ$. Self-membership is not prohibited in this framework: the Founding Axiom absorbs the self-referential loop as a terminal case rather than forbidding it as a paradox. This is the mathematical content of the remark that “ET absorbs Russell’s paradox”: the loop is the Exception’s defining feature, not a contradiction to be avoided.

Each Cardinal has a declared cardinality.

Axiom 2.5 (Cardinalities).

$$|P| = \Omega, \quad |D| = n, \quad |T| = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

where

- Ω is an *absolute infinity*: undifferentiated substrate, larger than every set-theoretic cardinal, not identifiable with any specific $\aleph_\kappa$;
- n is *absolutely finite*: a positive integer (variable across applications but always finite whenever D is bound to a specific P);
- $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ denotes *indeterminate* cardinality — neither finite nor infinite in the standard sense, but the cardinality of an operator whose outputs are determined by binding rather than by set-theoretic enumeration.

2.4 Three disjoint infinities

The declared cardinality $|D| = n$ characterises D *when bound to a specific P* ; unbound, the space of possible Descriptors is itself unbounded. Thus each of the three Cardinals is, in its own mode, infinite:

- P is *absolutely infinite*: undifferentiated substrate beneath any described content.
- D is *the infinite that chooses finitude*: the totality of possible constraints, unbounded when alone, rendered finite by the P – D singularity — the binding in which the infinite becomes small enough to be held. The finitude $|D| = n$ is the signature of that binding, not an inherent limit on D .
- T is *indeterminately infinite*: the cardinality $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is not the absence of cardinality, but the cardinality of genuine choice — the navigation of all possible configurations, which is, in its own mode, as total as either of the others.

These three infinities are categorically disjoint.

Axiom 2.6 (Categorical disjointness).

$$P \cap D = \emptyset, \quad D \cap T = \emptyset, \quad T \cap P = \emptyset.$$

2.5 Intrinsic mediation

The two preceding facts — that each Cardinal is infinite in its own mode, and that the three are pairwise disjoint — together force a structural consequence.

Proposition 2.7 (Intrinsic mediation). *If three Cardinals P, D, T are each unbounded in their respective modes and are pairwise disjoint (Axioms ??–??), then their coexistence within a single ontological space necessarily produces an internal binding relation. The relation is not a fourth primitive added from outside; it is the only structurally possible expression of three unbounded disjoint totalities coexisting.*

Argument. Three totalities each “filling all of its own mode of infinity” leave no exterior gap between them: a complete infinity, by definition, admits no external region in its own mode. Disjointness, however, demands that the three totalities remain categorically distinct. These two conditions are compatible only if the Cardinals mediate one another intrinsically — their interaction being the only possible way for three complete, disjoint infinities to coexist without contradiction. The operator $\circ$ of the master equation is this intrinsic mediation. It is not a postulated fourth element; it is a forced consequence of the three Cardinals’ joint presence. $\square$

$\square$

Remark 2.8. Proposition ?? removes a question that would otherwise need its own axiom: why is there a composition operator $\circ$ at all, and why is it ternary? The answer is that the existence and arity of $\circ$ are structural consequences of the Cardinals; $\circ$ does not need to be posited independently.

Remark 2.9 (Symmetry of mediation). The mediation is equal in all three Cardinals. P does not “receive” the mediation while D and T “perform” it; each Cardinal participates in $\circ$ with the same structural status. Any apparent asymmetry is an artefact of reading-order: the Cardinals are written $P \circ D \circ T$ for the binding-order reason of Axiom ??, but the binding itself is symmetric in that each Cardinal is equally a constituent of the Exception. Three complete disjoint infinities, by necessity, mediate each other; none of the three is merely the recipient of the other two.

2.6 The totality Σ

Definition 2.10 (Something). The *totality* Σ is

$$\Sigma = (P \circ D \circ T), \quad \forall x : x \in \Sigma.$$

Σ is constituted by *mediation*, not by *enumeration*. A set-theoretic union $P \cup D \cup T$ would combine the Cardinals without generative interaction, yielding no more than a list; the composition $P \circ D \circ T$ produces Something — the totality of configurations that can, or do, exist. This distinction will be essential: the ET lattice derived in the following sections is the discrete structure on the multiplicative aspect of Σ , not on a union of primitive sets.

Remark 2.11 (No outside). The statement $\forall x : x \in \Sigma$ asserts that nothing is outside Something; every object of any domain is a configuration of the three Cardinals. No alternative ontological category exists that is not already subsumed by P , D , or T . This claim is made rigorous by the Subsumption Law in the next section.

2.7 The binding order

Axiom 2.12 (Binding order). There is no configuration of the form $D \circ P$ preceding the establishment of P . Operationally, P must be specified before D can describe it; D must be specified before T can navigate it. The primitive ordering is $P \rightarrow D \rightarrow T$.

The force of Axiom ?? is structural. Any derivation from the Cardinals must respect the order: identify the substrate first, the constraints second, the agency third. Reversing the order collapses meaning. In the projection protocol of Section ?? this ordering appears concretely: the substrate is identified first (it determines the reference period), the lattice is the finite- D structure second, the rounding is the T -act third.

2.8 Modes of self-presentation

Each Cardinal reveals itself in a characteristic way; these modes are the diagnostic signatures by which P , D , T are identified in any concrete phenomenon.

- P reveals itself as *unreachable depth*. Any object has an apparent edge, but no edge terminates absolutely: beneath any described surface there is always more. The asymptotic limit that is approached but never attained is P . A pencil mark is the canonical illustration: the moment the point is “reached”, description has already been attached, and the bare Point recedes further.
- D reveals itself as the *articulable surface*. Everything that can be named, measured, formalised, or predicted is D -content. Mathematical laws, physical constants, sensory properties, chemical formulae, statistical distributions — all are members of D .

- T reveals itself as the *unrepeatable act of becoming*. Agency is visible only in the event of substantiation — not as a static property but as the act of resolving indeterminacy into an irreversible moment. Wavefunction collapse, the click of a detector, the decision of an agent, the rounding of a continuous quantity to a discrete lattice coordinate: these are T 's self-presentations. The rounding example is not chosen arbitrarily; it is the canonical operational enactment of T on the multiplicative manifold $(\mathbb{R}^+, \times)$, and reappears in §?? as the T -act of the projection formula.

These modes of self-presentation will be used implicitly when identifying which Cardinal is playing which role in a concrete example.

2.9 The four manifold states

The power set of $\{P, D, T\}$ has eight elements. Two are trivial (the empty configuration and the three singletons, which describe nothing by the master equation and binding order). Removing singletons and empties leaves exactly four states consisting of two or three Cardinals each.

Definition 2.13 (Manifold states). The four *manifold states* are

$$\{P, D, T\}, \quad \{P, D\}, \quad \{D, T\}, \quad \{P, T\},$$

called *Exception*, *Unsubstantiated*, *Mediation*, and *Incoherence* respectively.

We denote the cardinality of this set by S :

$$S = \binom{3}{2} + \binom{3}{3} = 3 + 1 = 4. \quad (1)$$

Thus $|P| = \Omega$, $|D| = n$, $|T| = [0/0]$, $|\Pi| = 3$, $S = 4$.

Proposition 2.14 (Incoherence is structurally forbidden). *The state $\{P, T\}$ admits no consistent binding: it contains substrate (P) and agency (T) but no finite constraint (D) by which agency can bind to substrate. Any attempt at a P – T binding produces a contradiction or dissolves into one of the other three states.*

The proof is immediate: without D , an element of T has no finite profile against which to test consistency with an element of P . In physical applications the $\{P, T\}$ state corresponds to null-pointer-like conditions: singularities, divisions by zero, paradoxes of self-reference without adequate stratification.

The three coherent states $\{P, D, T\}$, $\{P, D\}$, $\{D, T\}$ together with the forbidden state $\{P, T\}$ give the manifold Σ a precise topological structure.

Proposition 2.15 (Topological characterisation of the manifold states). *Let I denote the set of incoherent configurations $\{P, T\}$ and let ∂I denote its boundary. Then*

- (i) *The Exception $\{P, D, T\}$ is a closed set: $\partial E \subseteq E$. It contains its own ground.*
- (ii) *The Incoherent state $I = \{P, T\}$ is an open set: $\partial I \cap I = \emptyset$. It does not contain its own boundary.*
- (iii) *The Mediation state $\{D, T\}$ and the Unsubstantiated state $\{P, D\}$ are neither open nor closed; they are the transitional interior of the traversable manifold.*
- (iv) *The traversable manifold is $\Sigma \setminus I$, bounded from above by the closed E (the ground) and from the side by the open I (the edge).*

Definition 2.16 (The coherence–incoherence boundary ∂I). The *boundary* ∂I is the set of configurations at the limit of substantiability:

$$\partial I = \left\{ c \in \Sigma \mid \exists t \in T : \phi(t, c) = \lim_{\varepsilon \rightarrow 0} \phi(t, c_\varepsilon) \text{ where } c_\varepsilon \rightarrow c \right\},$$

where $\phi : T \times \Sigma \rightarrow \{0, 1\}$ is the substantiation indicator returning 1 when the triple binds into an Exception and 0 otherwise.

The boundary ∂I is the locus at which an arbitrarily small perturbation of the Descriptor content switches ϕ from 1 to 0. Physically it corresponds to second-order phase transitions, classical turning points, and marginal configurations in critical phenomena; mathematically it will reappear in Section ?? as the locus at which the lattice-adaptive fractal iteration transitions between polynomial-degree regimes, and every coherent configuration approaches ∂I arbitrarily closely but never crosses it from within coherence.

2.10 Variance and the grounding function

The distinction between the Exception and the other coherent configurations of Σ admits a direct formal expression as a function on the manifold.

Definition 2.17 (Variance function). The *variance function* at base resolution is

$$V(c) = \left| \left\{ c' \in \Sigma \setminus I \mid c' \neq c, \exists t \in \mathbb{T} \text{ such that } T(c, t) = c' \right\} \right|,$$

the cardinality of coherent configurations reachable from c by a single T -act. At base resolution V is non-negative and integer-valued. $V(c)$ measures c 's *capacity to be otherwise*: the range of alternatives that T can substantiate from c .

Remark 2.18 (Integrative-level refinement of V). The integer-valued form of Definition ?? is the base-resolution specification. At finer integrative levels (§??), the reachable-configuration count admits a continuous extension: V takes fractional values when the configuration space is given a continuous measure. The discrete (integer) and continuous (fractional) readings coincide on the invariant statements below and, as shown in the base-variance entry, on the numerical value of the minimum non-zero departure from E .

Proposition 2.19 (Zero variance characterises the Exception). *The Exception $E = P \circ D \circ T$ is the unique coherent configuration with zero variance:*

$$V(c) = 0 \iff c = E.$$

Proof. ($\Leftarrow$) By Axiom ??, E is the terminus of the exception-iteration while it IS: the distinguished configuration that admits no further exception. By Axiom ??, E is the fully substantiated binding of all three Cardinals; while E IS, no alternative coherent configuration is simultaneously the case, because any such alternative would itself be a distinct Exception at a distinct moment and therefore has already displaced this one. The set of coherent configurations reachable from E as E is empty, and $V(E) = 0$.

($\Rightarrow$) Let $c \in \Sigma \setminus I$ with $V(c) = 0$. By Definition ?? the coherent states are $\{P, D, T\}$, $\{P, D\}$, and $\{D, T\}$. If $c = \{P, D\}$ (Unsubstantiated), a T -act binding a Traverser to c produces $\{P, D, T\} = E$; if $c = \{D, T\}$ (Mediation), a T -act binding a specific P to c produces $\{P, D, T\} = E$; in both cases at least one alternative configuration is reachable, contradicting $V(c) = 0$. Hence $c \in \{P, D, T\}$. Uniqueness follows from the single-terminus clause of Axiom ??, formalised using T -time as $\exists! E_\tau$ in Proposition ?? of §??. $\square$ $\square$

Remark 2.20 (Variance is measured *from* the Exception). Proposition ?? is the ontological ground of variance as a measurement. Without the $V(c) = 0$ anchor supplied by E , there is no zero relative to which variance could be measured; a scale without a zero is not a scale. V is therefore not an independent primitive of Σ but a derived scalar whose existence and normalisation are consequences of the Founding Axiom and the master equation. *All positive variance in Σ is, by construction, variance from the Exception* — the $V > 0$ of non-Exception configurations is the direct expression of their departure from the unique grounded zero.

Ancillary variance forms and their relations to Definition ??

Several quantities also called “variance” appear in specific parts of the framework and its applications. Each stands in an explicit relation to $V(c)$ and to the manifold’s base quantum V_{base} .

Statistical variance of a discrete uniform descriptor distribution. If a Descriptor bound to a configuration takes one of n equally-weighted values $\{0, 1, \dots, n-1\}$, its classical statistical variance is

$$\sigma_{\text{disc}}^2(n) = \frac{n^2 - 1}{12}. \quad (2)$$

Equation (??) is not a restatement of $V(c)$: it is the second moment of the Descriptor distribution *within* a configuration’s D -content, not the cardinality of reachable configurations across Σ . The two are distinct quantities.

Statistical variance of a continuous uniform distribution on $[0, 1]$. The continuous analogue is the classical result

$$\sigma_{\text{cont}}^2 = \int_0^1 (x - \tfrac{1}{2})^2 dx = \frac{1}{12}. \quad (3)$$

Relation between (??) and (??). Rescaling the discrete uniform on $\{0, \dots, n-1\}$ by the factor $(n-1)$ so the support becomes $[0, 1]$ gives the normalised discrete variance

$$\sigma_{\text{norm}}^2(n) = \frac{n^2 - 1}{12(n-1)^2} \xrightarrow{n \rightarrow \infty} \frac{1}{12}, \quad (4)$$

which converges quantitatively to (??).

Asymptotic descriptor-count variance. In the regime of many independent Descriptors bound to a single Point, a per-descriptor variance quantity appears as

$$\text{Var}(D_n \rightarrow P) = \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0 \quad (\text{never reaches } 0). \quad (5)$$

Equation (??) expresses approach toward P -substrate dominance as the descriptor count grows. The limit is 0, but the limit is unattained — $V(c) = 0$ remains the exclusive signature of E (Proposition ??).

Remark 2.21 (The base variance $V_{\text{base}} = 1/N$). The *base variance* that appears among the three manifold constants (§??, Eq. (??)) is

$$V_{\text{base}} = \frac{1}{N},$$

with N the manifold symmetry forced to 12 in §??. Two complementary readings agree on this value.

- **Discrete reading (R1).** The minimum non-zero value of the integer-valued $V(c)$ of Definition ?? is 1: at least one reachable coherent configuration. Normalised by the manifold's N -fold symmetry, this minimum becomes $1/N$.
- **Continuous reading (R2).** At the continuous integrative level (Remark ??), V takes fractional values. The minimum positive value of the continuous extension at base resolution equals the continuous-uniform variance of Eq. (??), $\sigma_{\text{cont}}^2 = 1/12 = 1/N$.

Both readings coincide numerically at $V_{\text{base}} = 1/N = 1/12$. They are complementary specifications of the same scalar at different integrative levels, not competing interpretations. Equation (??) at $n = N$ and Eq. (??) in the limit likewise take the same value, providing further confirmation of the base quantum.

Remark 2.22 (Open question: sub-base quantisation). Whether V can take non-zero values strictly below $1/N$ is an open question. It has not been explored and is not derivable from what is currently established. It is flagged here as a direction for future work.

2.11 The four manifold states as four classical geometries

The four manifold states of Definition ?? are not purely combinatorial. They carry intrinsic geometric content inherited from the primitives' own operational manifolds. The mapping of states to classical geometries is forced, not chosen, by the geometry of those underlying manifolds.

Proposition 2.23 (Geometry of the primitives' own manifolds). *D 's operational domain $(\mathbb{R}^+, \times)$ under the logarithmic metric $ds^2 = dr^2/r^2 = d(\log r)^2$ has zero Gaussian curvature at every point. T 's operational domain $(U(1), \times)$ under its canonical bi-invariant metric has constant positive curvature $K_{U(1)} = 1/R^2$ for the unit circle. P has no intrinsic metric geometry: as Absolute Infinity, it carries no structure until D binds to it.*

Proof. The logarithmic metric on $\mathbb{R}^+$ pulls back via $\log_2$ to the flat Euclidean metric on $\mathbb{R}$; flat-manifold curvature is identically zero. The circle $U(1)$ with its bi-invariant metric is a compact 1-manifold of constant positive sectional curvature, standardly. P 's featurelessness (cardinality Ω , no intrinsic D -content) is axiomatic. $\square$

Corollary 2.24 (The full multiplicative manifold). *$\mathbb{C}^\times = (\mathbb{R}^+, \times) \times (U(1), \times)$ is a product of a flat manifold with a positively curved one, so it admits a canonical decomposition into a real axis (flat) and an angular axis (positively curved).*

This decomposition is what makes the real/imaginary projection formulas of §?? and §?? geometrically natural rather than stipulative: the real axis carries D 's flat geometry; the imaginary axis carries T 's positive curvature. Off-axis configurations interpolate. The resulting geometry-of-the-whole depends on which Cardinals are active, giving:

Proposition 2.25 (The four-states four-geometries mapping). *Under the identifications of Proposition ??, the four manifold states of Definition ?? correspond one-to-one with the four classical geometry types:*

State	Cardinals	Geometry type	Gaussian curvature K
<i>Exception</i>	$\{P, D, T\}$	<i>Euclidean</i>	$K = 0$
<i>Unsubstantiated</i>	$\{P, D\}$	<i>Elliptic</i>	$K > 0$ (constant)
<i>Mediation</i>	$\{D, T\}$	<i>Hyperbolic</i>	$K < 0$ (constant)
<i>Incoherence</i>	$\{P, T\}$	<i>Singular</i>	$K \rightarrow \pm\infty$

Proof. Exception $\{P, D, T\}$ requires P , D , and T all present and mutually balanced; in this configuration the base variance $V = 1/N$ is fully realised, giving a fully substantiated local configuration whose geometry is that of D 's flat manifold at the resolution chosen. $K = 0$ follows from Proposition ??.

Unsubstantiated $\{P, D\}$ has full D -content but no T -agency to navigate unbounded extensions; the geometry closes on itself, giving a compact positively curved surface. The canonical realisation is the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$; see Proposition ??.

Mediation $\{D, T\}$ has T operating without a fixed P -substrate to anchor it; parallel D -paths diverge under T 's navigation; this is the canonical hyperbolic configuration.

Incoherence $\{P, T\}$ has T attempting to bind P without D 's finite mediation (Proposition ??): the local metric becomes undefined. Geometrically this is the singularity condition $K \rightarrow \pm\infty$. $\square$

Proposition 2.26 (The Sempaevum's shape is state-dependent). *The Sempaevum does not have a fixed shape independent of the manifold state it is currently rendering. Under state transition $s_1 \rightarrow s_2$ within $\{\text{Exception}, \text{Unsubstantiated}, \text{Mediation}, \text{Incoherence}\}$, the Sempaevum's local geometry transitions simultaneously between the corresponding entries of Proposition ??. The refinement tower of §?? classifies all four state-geometries uniformly; no separate tower is needed for each.*

Proof. The projection formula $k = \text{round}(N \log_2 r)$ of §?? is independent of curvature. Curvature enters only through the choice of reference period R_0 used to form r , which is determined by the active manifold state (Three Tools, §??). Under state transition, R_0 transitions in lock-step with the geometry; r rescales to match; the projection formula evaluates on the new ratio. The sublattice family classification is unchanged. The Sempaevum's representation therefore tracks the shape without needing a separate apparatus for each geometry. $\square$

Proposition 2.27 (The Riemann sphere as the Sempaevum's $\{P, D\}$ configuration). *The one-point compactification $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ — the Riemann sphere — is the canonical elliptic form of the Sempaevum, arising under the $\{P, D\}$ Unsubstantiated manifold state on the complex multiplicative axis. Its symmetry group is $\text{PSL}(2, \mathbb{C})$, which is isomorphic to the identity component of the Lorentz group $\text{SO}^+(3, 1)$.*

Proof. $\hat{\mathbb{C}}$ is a compact positively curved surface with $\chi = 2$. Its biholomorphism group is the Möbius group $\text{PSL}(2, \mathbb{C})$, which acts transitively on triples of points. The isomorphism $\text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ is a classical result [?]: a Möbius transformation lifts to a conformal transformation of S^2 , which in turn corresponds to a proper orthochronous Lorentz transformation via the identification of S^2 with the celestial sphere of Minkowski space. The conditions for the $\{P, D\}$ state — full D -content, no T -navigation beyond the compact closure — are exactly the conditions for compactification to $\hat{\mathbb{C}}$. $\square$

Remark 2.28 (The four classical surface types as four states on one Sempaevum). The familiar classification of closed 2-manifolds by Euler characteristic χ — sphere ($\chi = 2$), torus ($\chi = 0$), genus- g surface ($\chi = 2 - 2g$), and singular boundary (χ ill-defined) — maps onto the four manifold states via Gauss–Bonnet (§??, Proposition ??). Sphere corresponds to $\{P, D\}$ Unsubstantiated with $d = 3$ under the Sempaevum's sublattice classification; torus corresponds to $\{D, T\}$ Mediation at $\chi = 0$, $d = 1$ (the zero-total-curvature / octave attractor); higher-genus hyperbolic surfaces correspond to $\{D, T\}$ at variable d ; singular configurations correspond to $\{P, T\}$ Incoherence. Lattice, torus, Riemann sphere, and genus- g hyperbolic surface are not separate entities in the Sempaevum framework: they are the Sempaevum wearing its four different state-geometries.

3 The triple identity and the three operational tools

3.1 Three simultaneous categorical readings

The structure given in §?? admits three parallel categorical readings, each complete, each with exactly three members, each mutually entailing the other two.

Theorem 3.1 (The triple identity $3 = 3 = 3 = \Sigma$). *The three Cardinals admit three complete simultaneous readings whose joint identity is*

$$\boxed{\text{PDT} = \text{EIM} = \Phi = \Sigma,}$$

or in compact form $3 = 3 = 3 = \Sigma$, where

- $\text{PDT} = \{P, D, T\}$ is the ontological triad: what each Cardinal is;
- $\text{EIM} = \{E, I, M\}$ is the phenomenological triad: the Exception E (grounded actuality), Incoherence I (the coherence boundary as a structural fact), Mediation M (the binding operation in action);
- $\Phi = \{\Phi_E, \Phi_I, \Phi_M\}$ is the impossibility triad: $\Phi_E =$ cannot be otherwise, $\Phi_I =$ cannot be traversed to, $\Phi_M =$ cannot be absent.

All three triads read the same structure from different angles. None is more fundamental than the others; each entails both of the others.

The correspondence between positions is summarised in Table ??.

Position	PDT (ontological)	EIM (phenomenological)	Φ (impossibilities)
First	P — substrate, Ω	E — grounding	cannot be otherwise
Second	D — constraint, n	I — coherence boundary	cannot be traversed to
Third	T — agency, $[0/0]$	M — mediation	cannot be absent
View	structural (outside)	experiential (inside)	boundary (forbidden)
Answers	<i>What is it?</i>	<i>What does it give?</i>	<i>What can never be?</i>

Table 1: The triple identity $3 = 3 = 3 = \Sigma$. Each row is a single structural fact read three ways. Cardinalities carry across readings: $|E| = \Omega$, $|I| = n$, $|M| = [0/0]$, because E is what P gives, I is what D enforces, M is what T is.

Derivation of any triad from any other proceeds directly from the table. From PDT alone: P is substrate of cardinality $\Omega \Rightarrow$ its contribution is grounding $E \Rightarrow$ the ground cannot be otherwise (Φ_E). Similarly $D \rightarrow I \rightarrow \Phi_I$ and $T \rightarrow M \rightarrow \Phi_M$.

3.2 The universality anchor $3 = 3 = 3 = \Sigma$ versus the composition equation $P \circ D \circ T = E$

Two distinct identities are frequently conflated and must be held apart.

Axiom 3.2 (The composition equation, restated). When the three Cardinals bind into a single configuration, an Exception is produced:

$$P \circ D \circ T = E.$$

Axiom 3.3 (The universality anchor). The three complete readings of (P, D, T) co-referentially constitute the totality:

$$\text{PDT} = \text{EIM} = \Phi = \Sigma, \quad \text{i.e.} \quad 3 = 3 = 3 = \Sigma,$$

with Σ the set-theoretic totality of everything that exists, could-exist, or is constrainedly-forbidden.

Remark 3.4 (When to use which). The composition equation $P \circ D \circ T = E$ produces substantiated moments: it addresses *one* of the four manifold states (the Exception). The universality anchor $3 = 3 = 3 = \Sigma$ addresses the totality: it covers all four manifold states simultaneously, including the Mediation $\{D, T\}$ transitions in progress, the Unsubstantiated $\{P, D\}$ structures that have not been traversed (including fiction, hypotheticals, non-computable-but-definable reals), and even the Incoherent $\{P, T\}$ configurations (which sit at the boundary ∂I , hence *on* the structure, not outside it). For the question “is X lattice-addressable?” the correct anchor is $3 = 3 = 3 = \Sigma$, since $X \in \Sigma$ for any X describable at all; $P \circ D \circ T = E$ alone would restrict address-ability to fully-substantiated objects.

Theorem 3.5 (Identification: the Sempaevum is Σ rendered mathematically). *The multiplicative lattice structure on $(\mathbb{R}^+, \times) \times (U(1), \times)$ constructed in §§??–?? is identified with Σ . In precise terms, it is the unique object in the category of “mathematical totalities” that passes all three Subsumption Law tests (§??) simultaneously:*

- (i) *It cannot be subsumed by any other candidate (no proper subset of its closure properties is itself closed).*
- (ii) *Nothing external subsumes it (every proposed larger structure either collapses into it or reproduces its defining primitives).*
- (iii) *It subsumes every mathematical totality-candidate without remainder (every conventional mathematical universe is recoverable as a restriction or rendering, proved case-by-case in §??).*

The identification is a theorem of Σ ’s uniqueness under the Subsumption Law; it is not an assertion. This object we call the SEMPAEVUM.

The proof occupies the rest of the paper: every subsequent theorem is a property that a candidate identification of Σ must satisfy, and each is established forward from (P, D, T) without external axioms.

3.3 The three operational tools

The axiom system of §?? supports three operational tools that together form the complete methodology of Exception Theory. Each is a direct consequence of the Founding Axiom and the master equation; none is independently postulated.

- The **Identification Principle** specifies *what to find*: the complete P – D – T decomposition of the phenomenon under study.
- The **Descriptor Gap Principle** specifies *how to find it*: every gap in a model is itself a Descriptor not yet identified.
- The **Subsumption Law** specifies *when the process terminates*: the description is complete when it subsumes the phenomenon without remainder.

The three are presented formally below. They form a closed convergent loop (§??) that is both the method by which the ET framework was itself discovered and the method by which, in Sections ??–??, the lattice is derived and confirmed.

3.4 The Identification Principle

Theorem 3.6 (Identification Principle). *For any phenomenon X , understanding of X is complete if and only if all three of its PDT components are identified:*

$$\boxed{\text{Understand}(X) \iff \text{Identified}(P_X) \wedge \text{Identified}(D_X) \wedge \text{Identified}(T_X).}$$

Argument. By the master equation (Axiom ??), any substantiated configuration is of the form $P \circ D \circ T = E$. By Definition ??, the four manifold states partition substantiable configurations: only the Exception $\{P, D, T\}$ is fully substantiated. If any of P_X , D_X , T_X is unidentified, the model of X occupies one of the three non-Exception states (Unsubstantiated, Mediation, or Incoherence) and is an incomplete description. Conversely, if all three are identified, the configuration is the Exception and the model is complete. The biconditional follows. $\square$ $\square$

The Identification Principle inherits the binding order of Axiom ??: operationally, P_X must be identified first, D_X second, T_X third. An under-specified P_X leaves D and T with nothing to attach to; an over-specified P_X smuggles D -content into the substrate and produces false confidence. The three diagnostic questions, in order, are:

- (i) What is the substrate? (seeks P_X)
- (ii) What are the constraints? (seeks D_X)
- (iii) What agency navigates through them? (seeks T_X)

If any question cannot be answered, understanding is incomplete, and the unanswered question identifies exactly which Cardinal is missing.

Remark 3.7 (Diagnostic power). When a scientific model fails — when predictions diverge from measurement or the mathematics becomes inconsistent — the first-order question of Exception Theory is not “what additional equations are needed?” but “which of the three Cardinals has not been properly identified?”. This converts vague confusion into a specific, actionable search target chosen from exactly three possibilities.

Remark 3.8 (Application to the lattice). The derivation of $N = 12$ in Section ?? is itself an application of the Identification Principle to the framework of the lattice. The substrate of the lattice is $P_{\text{lat}} = \mathbb{R}^+$ under multiplication (the space of all positive ratios). The Descriptor of the lattice is $D_{\text{lat}} = \Pi \cup \text{States} = \{P, D, T\} \cup \{\{P, D, T\}, \{P, D\}, \{D, T\}, \{P, T\}\}$, supplying the two finite counts $|\Pi| = 3$ and $|\text{States}| = 4$. The Traverser of the lattice is $T_{\text{lat}} =$ the rounding operator that projects a continuous ratio onto a discrete lattice coordinate. The unique integer that emerges from this identification is $N = 3 \cdot 4 = 12$. This derivation is worked out in full in §??.

3.5 The Descriptor Gap Principle

The second tool converts any gap in an incomplete description into a well-posed search target.

Theorem 3.9 (Descriptor Gap Principle). *For any model $\mathcal{M}$ of a phenomenon X ,*

$$\boxed{\text{gap}(\mathcal{M}) \in D_{\text{missing}}.}$$

Every gap in a model is itself a Descriptor that has not yet been identified.

Argument. A gap in $\mathcal{M}$ is a specific characterisation of $\mathcal{M}$: the characterisation “ $\mathcal{M}$ is incomplete in the following respect”. By Definition ?? and Proposition ??, the category of all characterisations, constraints, and rules is D . The gap is therefore an element of D . Since, by hypothesis, the gap has not been captured by the Descriptor set of $\mathcal{M}$, it is an element of D not yet attached to the model: $\text{gap}(\mathcal{M}) \in D_{\text{missing}}$. $\square$ $\square$

Corollary 3.10 (Verification Principle). *The mathematical consistency of a model is equivalent to the absence of Descriptor gaps:*

$$\text{Consistent}(\mathcal{M}) \iff D_{\text{missing}}(\mathcal{M}) = \emptyset.$$

The Descriptor Gap Principle is the formal content of the methodological claim that “any problem can be solved — it is a matter of the right Descriptors and the number of Descriptors”. A failing model is not an obstacle but a pointer to its own completion: the failure is a gap; the gap is a Descriptor; the Descriptor is to be found. The process converges because each identified Descriptor is a genuine ontological constraint not previously represented; no cycle is possible.

Remark 3.11 (Gödel and the infinite–finite asymmetry). The Descriptor Gap Principle is compatible with Gödel’s incompleteness theorems. A formal system expressive enough to interpret $\mathbb{N}$ admits undecidable statements G ; in the present language, such a G is the manifold’s expression of the fact that $|P| = \Omega$ cannot be exhausted by any finite D -set of size n . The gap is structural, and it is itself a Descriptor: the statement “this system’s finite D -structure cannot capture all truths about its Ω -substrate”. Gödel’s theorem does not falsify Exception Theory; it is the expected signature of $|P| = \Omega \gg n = |D|$ at every finite level of description.

Remark 3.12 (Application to the lattice). The four-term identity for α^{-1} established in §?? (equation (??)) is a closure of a long-standing Descriptor gap. The deviation of the measured fine-structure constant from the integer 137 was, historically, an unexplained fraction. The Descriptor Gap Principle recasts the deviation as a call for Descriptors: a $+\sqrt{3}/48$ shimmer contribution from P – D – T cross-interactions, a $-\sqrt{3}/(93312\pi^2)$ shimmer-loop correction, and a $-1/(216(18\pi-1))$ closed-mediation-loop correction. With these three Descriptors attached, the model reproduces the measured value to sub-ppb precision. Each Descriptor was a gap made explicit.

3.6 The Subsumption Law

The third tool is the operational criterion that validates the three Cardinals as complete — and, more generally, tells one *when* a decomposition is finished.

Principle 3.13 (Subsumption Law). A primitive Cardinal is complete and irreducible if and only if it satisfies all three of the following conditions:

- (i) It cannot be subsumed by either of the other two Cardinals.
- (ii) Nothing external to the three Cardinals subsumes it.
- (iii) It subsumes everything within its own category without remainder.

The system $\{P, D, T\}$ is complete if and only if every Cardinal in it passes all three conditions.

The Subsumption Law is the primary completeness criterion of Exception Theory. It is an empirical-structural test rather than an additional postulate: the three Cardinals are declared complete because they pass the test, and a fourth primitive is ruled out because no candidate fourth can.

Proposition 3.14 (Irreducibility of P, D, T). *The three Cardinals P, D, T satisfy the Subsumption Law.*

Argument. We verify condition (i) for each Cardinal; conditions (ii) and (iii) follow by the exhaustive trichotomy of modes of infinity.

- P cannot reduce to D (a constraint requires something to constrain; substrate cannot be its own constraint) or to T (substrate has no internal direction of choice).

- D cannot reduce to P (undifferentiated substrate supplies no structure) or to T (constraints are deterministic; agency is indeterminate).
- T cannot reduce to P (substrate has no agency) or to D (fixed rules cannot choose which rule to follow).

All three conditions are met for all three Cardinals. $\square$ $\square$

Corollary 3.15 (No fourth primitive). *No Cardinal X outside $\{P, D, T\}$ is a primitive.*

Argument. Any candidate X falls into one of four cases: (a) X has Ω -type infinity, hence $X \subseteq P$; (b) X is finitely constraining, hence $X \subseteq D$; (c) X is agential-indeterminate, hence $X \subseteq T$; (d) X lies outside all three categories. In case (d), by Subsumption Law condition (iii) applied to each of P, D, T , X has no instance in Σ and is therefore not a primitive. In cases (a)–(c), X is subsumed by one of P, D, T and so is not a new primitive. $\square$ $\square$

The Subsumption Law will reappear in §?? as the structural reason for the universality of the lattice: every positive-ratio quantity in every domain lies in Σ , and Σ is exhausted by configurations of P, D, T , so every positive-ratio quantity admits a lattice projection.

3.7 The closed methodological loop

The three tools of §§??–?? are not independent heuristics; they are mutually interlocking, and together form a closed, convergent procedure.

Proposition 3.16 (The three-tools loop). *Given a phenomenon X and an initial model $\mathcal{M}_0$, the procedure*

- (1) *Identify: apply Theorem ?? to diagnose which of P_X, D_X, T_X is missing in $\mathcal{M}_k$;*
- (2) *Gap: apply Theorem ?? to convert the missing component into a specific Descriptor to be found;*
- (3) *Verify: test the updated model $\mathcal{M}_{k+1}$ by Corollary ??;*
- (4) *Subsume: apply Principle ?? to check whether the description subsumes X without remainder;*

terminates in finitely many steps at a model $\mathcal{M}_K$ which is mathematically consistent and subsumption-complete, or else proves that X is not expressible within Σ (in which case X does not exist as a substantiated configuration).

The sketch of the proof is that each iteration either (i) terminates with Subsumption achieved, or (ii) adds a genuinely new Descriptor to the model. Since each added Descriptor is a new element of D not previously identified, and since D has finite cardinality n once bound to a specific P_X , the sequence of iterations cannot cycle and must terminate after at most n steps. Non-existence of a terminating configuration would place X outside the totality Σ , contradicting Definition ??.

Remark 3.17 (The loop applied to ET itself). Exception Theory was discovered by iterating the three-tool loop on the question “what is the minimal complete description of reality?”. The Identification Principle pointed at substrate, constraint, and agency as the three irreducible categories; the Descriptor Gap Principle refined each category through successive applications (resolving gaps such as the nature of time, the hydrogen atom, consciousness); the Subsumption Law terminated the recursion at three primitives by ruling out a fourth. The framework presented in §?? is the Subsumption-complete output of this iterative process.

Remark 3.18 (The loop applied to the lattice). The lattice of Section ?? is likewise the output of a three-tools iteration: identifying (P, D, T) of the multiplicative manifold, closing gaps (N, V, K , the α^{-1} corrections, the self-consistency attractor), and verifying subsumption across the full range of empirical domains canvassed in §??.

3.8 Integrative levels

A remaining structural observation concerns the Cardinals' relation to the hierarchy of physical and mathematical phenomena.

Definition 3.19 (Integrative level). An *integrative level* is a level of organisation at which a collection of Cardinal configurations constitutes a whole whose properties are not present or derivable at lower levels.

The quantum, atomic, molecular, cellular, biological, cognitive, and civilisational levels are examples. At each, properties emerge (wavefunction superposition, chemical bonding, molecular reactivity, life, consciousness, language) that are not possessed by, and are not obtainable by summing, the Cardinals of the constituents below.

Proposition 3.20 (Governance at all integrative levels). *The Cardinals P, D, T govern every integrative level without modification. Emergent properties at higher levels are new configurations of the same three Cardinals; no additional Cardinal is introduced by complexity.*

The proof is by the Subsumption Law: any property emerging at a higher integrative level must be substrate-like (P -type), constraint-like (D -type), or agency-like (T -type). Nothing outside the three categories appears at higher levels. What changes across integrative levels is not the Cardinals but the identification depth at which the master equation is applied, and the number and complexity of Descriptors required.

The mathematical content of wholeness is the set-of-all-sets structure of Definition ??; because each Cardinal is the totality of its kind, combining configurations into a whole contributes real mathematical content (the wholeness relation itself), producing properties that are not present at the level of constituents. Emergence, in this framework, is not an unexplained phenomenon but a direct consequence of the Cardinal structure.

The identification depth referenced above is not merely qualitative. It has a precise quantitative counterpart in the lattice itself: the LCM tower of refinements in Section ?. Each level of the tower $N \in \{12, 60, 420, 2520, 27720, \dots\}$ makes a specific set of sublattice families d available as native divisors, and a phenomenon at integrative level ℓ is fully identifiable at resolution N if and only if every sublattice family appearing in its (P, D, T) decomposition is native at N . The qualitative progression *quantum* $\rightarrow$ *atomic* $\rightarrow$ *molecular* $\rightarrow$ *cellular* $\rightarrow$ *biological* $\rightarrow$ *cognitive* $\rightarrow$ *civilisational* has a quantitative shadow in the progression $12 \rightarrow 60 \rightarrow 420 \rightarrow 2520 \rightarrow 27720 \rightarrow \dots$. The correspondence is made precise by Proposition ??; the specific mapping of empirical phenomena to resolution levels appears in Table ??; and the exact quantitative law relating the two ladders is the doubling equation $\tau(N_\ell) = 6 \cdot 2^\ell$ of Theorem ??, which states that each integrative emergence doubles the number of sublattice families available — a number-theoretic fact, a phenomenological fact, and the same fact.

The ET lattice is therefore not blind to integrative level but actively encodes it: a given ratio r projects to the same lattice coordinate (k, d, ε) regardless of whether it originates in atomic mass ratios, musical intervals, biological symmetries, economic doublings, or civilisational cycles, and its sublattice family d identifies which integrative level's resolution is minimally required to host it. The lattice simultaneously classifies across integrative levels (through d) and operates uniformly within them (through the projection formula).

4 The three temporal aspects

Time is one of the canonical domains to which the Identification Principle of §?? applies. Classical physics carries two distinct notions of time — coordinate time t and proper time τ — without a unifying substrate. The Identification Principle, applied to time, identifies these two as D -time and T -time respectively and forces the recognition of a third temporal aspect, P -time, as the pre-geometric temporal substrate.

4.1 The three temporal aspects

Definition 4.1 (Temporal aspects). Applying the master equation $P \circ D \circ T = E$ to the temporal domain identifies three temporal aspects:

Aspect	Cardinality	Physics analog	Nature
P_{time}	Ω	pre-geometric temporal field	undifferentiated, featureless container of temporal po
D_{time}	n	coordinate time t	finite ordering descriptor; imposes sequence and direc
T_{time}	$[0/0]$	proper time τ	accumulated substantiation history of a Traverser

P_{time} is the undifferentiated infinite container of temporal potential: every temporal slot is identical to every other before D_{time} makes contact. Temporal ordering — past-to-future, before-and-after — is a D -structure applied to P_{time} , not an intrinsic property of the substrate. D_{time} is a finite ordering descriptor that imposes sequence on P_{time} configurations; in physics it corresponds to coordinate time t . T_{time} is local and path-dependent: each Traverser has its own T_{time} , given by the accumulated count of substantiation events along the Traverser’s worldline. It corresponds to proper time τ in physics.

The categorical disjointness and intrinsic mediation established for the general Cardinals (§??–§??) specialise to time without modification: the three temporal aspects are categorically disjoint and mediate one another intrinsically, just as P , D , T do in general.

4.2 The temporal master equation

Axiom 4.2 (Temporal master equation).

$$P_{\text{time}} \circ D_{\text{time}} \circ T_{\text{time}} = E_{\text{moment}}.$$

E_{moment} is the temporal Exception — the grounded, irreversible *Now*. It is the specialisation of the general master equation (Axiom ??) to the temporal domain. Each substantiated moment of time is an instance of E_{moment} : all three temporal aspects bound, $V(E_{\text{moment}}) = 0$ by Proposition ??, and uniquely current in the sense made precise immediately below.

4.3 Uniqueness of the temporal Exception

Proposition 4.3 ($\exists! E_\tau$). *At each T -time value τ there is exactly one temporal Exception:*

$$\forall \tau : \quad \exists! E_\tau \in \Sigma.$$

Multiple Traversers may share E_τ — this is the structural basis of shared reality — but E_τ itself is unique at its T -time.

Proof. Suppose two distinct temporal Exceptions $E_{\tau,1}$ and $E_{\tau,2}$ are both maximally substantiated at a single T -time τ . By Axiom ??, the exception-iteration terminates at exactly one distinguished configuration — the Exception. Two distinct maximally-substantiated configurations at the same τ would each serve as an “exception” to the other (in the structural sense that each could be otherwise with respect to the other), and the iteration would fail to terminate at a single object. This contradicts the single-terminus clause of the Founding Axiom. Hence $E_{\tau,1} = E_{\tau,2}$. □

Proposition ?? closes the uniqueness clause invoked in the proof of Proposition ?? (§??): T -time τ is the index under which the Exception is unique, and $V(c) = 0$ singles out that unique E_τ at each τ .

4.4 The Minkowski interval

The separation between D -time and T -time, together with the existence of a maximum descriptor-gradient c , yields the Minkowski interval of special relativity as a structural consequence of the three-aspect decomposition.

Proposition 4.4 (Minkowski interval). *Let a Traverser navigate an interval dt of D -time accompanied by a spatial descriptor shift dx . Let $v = dx/dt$ denote the spatial velocity and c the maximum descriptor-gradient ratio of the manifold. Then the T -time increment $d\tau$ satisfies*

$$d\tau^2 = dt^2 - \frac{dx^2}{c^2}$$

equivalently $d\tau = dt\sqrt{1 - v^2/c^2}$.

Proof. A Traverser's total traversal capacity per unit D -time is bounded by the manifold's maximum descriptor-gradient c . This capacity is partitioned between temporal substantiation, which accumulates T -time at rate $d\tau/dt$, and spatial descriptor shift, which accumulates position at rate $v = dx/dt$. The squared fractions sum to unity:

$$\left(\frac{d\tau}{dt}\right)^2 + \left(\frac{v}{c}\right)^2 = 1.$$

Rearranging,

$$d\tau^2 = dt^2 \left(1 - \frac{v^2}{c^2}\right) = dt^2 - \frac{dx^2}{c^2}. \quad \square$$

□

Remark 4.5 (Limiting cases). At $v = 0$ all of T 's capacity is in temporal substantiation and $d\tau = dt$. At $v = c$, T 's capacity is exhausted in the spatial direction and $d\tau = 0$ — the proper-time experience of a photon. Proposition ?? reproduces the special-relativistic time-dilation formula $d\tau = dt\sqrt{1 - v^2/c^2}$ from the three-aspect temporal decomposition alone; no independent postulate of Lorentz invariance is required.

4.5 Event-horizon reconciliation

The three-aspect decomposition resolves the apparent tension between two standard descriptions of matter falling into a Schwarzschild black hole: the external observer's report that matter freezes at the horizon, and the infalling observer's report of finite crossing time.

Let M denote the mass of a non-rotating uncharged black hole, and let $r_s = 2GM/c^2$ denote its Schwarzschild radius. Standard general relativity gives, for a static observer at radial coordinate $r > r_s$, the proper-time to coordinate-time ratio

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{r}}. \quad (6)$$

Proposition 4.6 (Event-horizon reconciliation). *Under the three-aspect temporal decomposition of Definition ??, the two standard descriptions of matter approaching the Schwarzschild radius are simultaneously valid:*

- (i) *For a static external observer parametrised by D -time, $d\tau/dt \rightarrow 0$ as $r \rightarrow r_s^+$; infalling matter appears frozen at the horizon in D -time.*
- (ii) *For a radially infalling observer parametrised by T -time, the T -time required to reach $r = r_s$ from any $r_0 > r_s$ is finite.*

The apparent tension between (i) and (ii) dissolves under the recognition that D -time and T -time are categorically distinct temporal aspects (Definition ??); neither is privileged and neither description falsifies the other.

Proof. (i) Equation (??) gives $d\tau/dt = \sqrt{1 - r_s/r}$; as $r \rightarrow r_s^+$ the ratio tends to zero, so $dt/d\tau \rightarrow \infty$. In the D -time frame, unbounded D -time is required to register any finite T -time advance near the horizon; this is the precise meaning of “coordinate freezing.”

(ii) For a radially infalling timelike geodesic from rest at $r_0 > r_s$, integration of the Schwarzschild radial geodesic equation in proper-time parametrisation gives the proper-time to reach $r = 0$ (the central singularity) as

$$\Delta\tau_{r_0 \rightarrow 0} = \frac{\pi}{2} \cdot \frac{r_0^{3/2}}{\sqrt{r_s} c^2},$$

which is finite for any finite r_0 . The proper time to reach $r = r_s$ from r_0 is bounded above by $\Delta\tau_{r_0 \rightarrow 0}$ and is therefore also finite. The T -time of the infalling Traverser at the horizon is finite.

Reconciliation. The freezing of (i) is a statement in D -time; the finite crossing of (ii) is a statement in T -time. By Definition ?? these are distinct temporal aspects; by Axiom ?? each is an irreducible component of the temporal master equation; neither can be reduced to the other. Both descriptions describe the same physical situation under two different temporal aspects, and both are correct. $\square$ $\square$

Remark 4.7 (Historical reading). Under the three-aspect temporal decomposition, Einstein’s classical statement of the horizon and the later recognition that geodesic completeness extends through it are not competing theories. Einstein’s statement is the D -time truth of Proposition ??(i); the extended-manifold statement is the T -time truth of Proposition ??(ii). The historical tension between them was an artefact of using a single, undifferentiated notion of “time” to compare two descriptions that operated in different temporal aspects.

4.6 Scope of this section

Two further physics reconciliations follow from the three-aspect temporal decomposition but are not established in this section: the Hawking temperature read as a time-ratio quantity at the horizon, and the black-hole information question. Both require additional mathematical treatment against standard quantum field theory in curved spacetime and are deferred to a dedicated later section; they are noted here to make the scope of the present section explicit. The Minkowski interval (§??) and the event-horizon reconciliation (§??) are structural consequences of the three-aspect decomposition alone and are established above. The material of this section — the three temporal aspects, the temporal master equation, the uniqueness $\exists! E_\tau$, the Minkowski interval, and the event-horizon reconciliation — is the structural foundation on which the deferred reconciliations will rest.

5 The forced value $N = 12$

The content of Section ?? is all of the machinery we shall use to derive $N = 12$. We give three independent derivations; each fixes the same integer.

5.1 Derivation I: product of the two counts

The two finite cardinals available in Axiom ?? and Definition ?? are

$$|\Pi| = 3 \quad (\text{number of primitive categories}), \quad S = 4 \quad (\text{number of non-trivial manifold states}).$$

Proposition 5.1 (Manifold symmetry). *The manifold symmetry is*

$$N = |\Pi| \cdot S = 3 \cdot 4 = 12.$$

The formula $N = |\Pi| \cdot S$ is the natural product of the independent counts: for each of the three primitive Cardinals, there are four structurally distinct manifold states to which a given element can contribute (Exception, Unsubstantiated, Mediation, or the forbidden Incoherence). The count is unique up to the ordering of the cardinalities fixed in Axiom ??.

5.2 Derivation II: least common multiple of the small sublattice orders

Suppose we seek the smallest positive integer N such that $\{1, 2, 3, 4\}$ all divide N ; these are the integer orders obtainable from the $\binom{4}{k}$ counts on a set of cardinality 4 (namely $\{1, 4, 6, 4, 1\}$ restricted to ≤ 4). The unique minimum is

$$\text{lcm}(1, 2, 3, 4) = 12.$$

Raising the requirement to $\{1, 2, 3, 4, 5\}$ gives $\text{lcm} = 60$; raising to $\{1, \dots, 7\}$ gives 420; raising to $\{1, \dots, 11\}$ gives 27720. The sequence 12, 60, 420, 2520, 27720 will reappear in Section ?? as the LCM tower of the ET lattice. The base value 12 is the smallest member of this tower: the minimal common multiple of the fundamental small divisors.

5.3 Derivation III: the stability window

Consider the discrete multiplicative group $\mathbb{Z}/N\mathbb{Z}$ generated by a unit g coprime to N . A necessary condition for the generator g to produce a complete cycle through $\mathbb{Z}/N\mathbb{Z}$ is $\gcd(g, N) = 1$. A natural additional condition, which we shall call the *stability window*, is that the multiplicative deviation accumulated per step on the corresponding logarithmic lattice $\{2^{k/N}\}_{k \in \mathbb{Z}}$ stay below an auditorily imperceptible threshold of 50 cents (half a lattice step):

$$N \cdot |\delta| < 50 \text{ cents}, \quad \delta = \log_2(3/2) - \frac{7}{12} \approx 0.0019550.$$

Here δ is the per-step discrepancy between the fifth generated by the circle-of-fifths walk $g = 7$ on $\mathbb{Z}/12\mathbb{Z}$ and the just fifth $3/2$. With $N = 12$ and $|\delta| = 0.0019550$ octaves, we have

$$N \cdot |\delta| \approx 12 \cdot 2.346 \text{ cents} \approx 23.5 \text{ cents} < 50 \text{ cents}.$$

For $N = 11$ the stability window condition fails against the nearest available generator; for $N = 13, 14, \dots$ the window is open, but the LCM-minimality condition of Derivation II forces $N = 12$. The manifold symmetry value $N = 12$ is the unique integer ≤ 60 that simultaneously satisfies:

- (i) the stability window $N \cdot |\delta| < 50$ cents against the circle-of-fifths generator $g = 7$;
- (ii) the LCM-minimality $N = \text{lcm}(1, 2, 3, 4)$;
- (iii) a maximal palindromic depth $\tau(12) = 6$ (the number of divisors of 12 is 6; no smaller integer attains this count).

5.4 The three manifold constants

The following constants are direct consequences of Proposition ?? and Definition ??:

$$\boxed{N = 12, \quad V = \frac{1}{N} = \frac{1}{12}, \quad K = \frac{2}{3}.} \tag{7}$$

Here V is the *base variance*: the minimum non-zero value of the variance function of Definition ??, equal to $1/N$ at the manifold's base resolution. The discrete and continuous readings of V coincide at this value (Remark ??); whether V can take non-zero values strictly below $1/N$ is an open question (Remark ??). With $N = 12$ forced by Proposition ??, $V_{\text{base}} = 1/12$ is the forced value of this quantum. The value K is derived from Definition ??: of the four manifold states, three contain P (Exception, Unsubstantiated, Incoherence) and two contain both P and D (Exception, Unsubstantiated); more tellingly,

$$K = 1 - \frac{S}{N} = 1 - \frac{4}{12} = \frac{2}{3},$$

which is the complement of the ratio of state count to manifold symmetry. The value $K = 2/3$ will reappear as the Koide stability threshold.

5.5 A derived fine-structure constant

Two counts can be combined further. Define

$$A_0 = (N - 1)^2 + S^2 = 11^2 + 4^2 = 121 + 16 = 137. \quad (8)$$

The quantity A_0 is the sum of the squares of the two available independent-count residues: $(N - 1)$ is the count of non-trivial logarithmic steps per octave (excluding the origin), and $S = 4$ is the count of manifold states. The identity $A_0 = 137$ is exact, derived, and of direct physical interest: the measured fine-structure constant in the Lattice Compendium 2025 reference is $\alpha^{-1} = 137.0359991\dots$, leaving a residual of 0.036 to be accounted for by higher-order corrections (Section ??).

6 The lattice on the multiplicative manifold

We now define the lattice and prove its convention-independence.

6.1 The projection formula

Definition 6.1 (ET real-axis projection). Let N be a positive integer (the *lattice resolution*, taken to be 12 unless otherwise stated), and let $r \in \mathbb{R}^+$ be a positive real. The *ET projection* of r at resolution N is the triple $(k, d, \varepsilon) \in \mathbb{Z} \times \mathbb{N} \times \mathbb{R}$ defined by

$$k = \text{round}(N \log_2 r), \quad (9)$$

$$g = \gcd(|k|, N), \quad (10)$$

$$d = N/g, \quad (11)$$

$$\varepsilon = (N \log_2 r - k) \cdot \frac{1200}{N} \text{ cents}, \quad (12)$$

with the convention $g = N$ (hence $d = 1$) when $k = 0$. We write the projection as $\Pi_N(r) = (k, d, \varepsilon)$.

We call k the *lattice coordinate*, d the *sublattice family*, and ε the *descriptor gap* (in cents, where one cent equals $1/100$ of a 12-ET semitone; the choice of $1200/N$ as the multiplier ensures that ε has units of cents at any N).

6.2 Interpretation

The lattice is the set

$$\mathcal{L}_N = \{ 2^{k/N} : k \in \mathbb{Z} \} \subset (\mathbb{R}^+, \times).$$

At $N = 12$ this is exactly the set of frequency ratios accessible on a twelve-tone equal-tempered scale, but the derivation given above shows that 12-division is not a cultural choice.

The formula (??) says: write r as a real number of logarithmic steps, and round to the nearest integer number of steps. The formula (??) says: classify the resulting integer k by how deeply it sits in the divisor lattice of N ; if k is coprime to N , then $d = N$ (fine resolution); if k divides N heavily, then d is small. The formula (??) records the residual, in cents, by which r fails to land exactly on the lattice.

Conceptually, the three formulas are the triptych of the binding operation $P \circ D \circ T$:

- r is the P -content (a featureless positive real);
- N and the lattice structure are the D -content (finite constraints);
- round is the T -act (resolution of a continuous quantity to a discrete one).

Remark 6.2 (The projection as the master equation at single-ratio scope). The output triple (k, d, ε) is itself the Exception of this projection: a fully substantiated, zero-variance configuration bound by all three Cardinals, with k the D -coordinate in $\mathcal{L}_N$, d its sublattice family, and ε the descriptor gap in cents. The rounding step realises the canonical mode of T -self-presentation anticipated in §??. Equivalently, each projection is the master equation

$$P \circ D \circ T = E$$

enacted at the level of a single ratio: (k, d, ε) is the structural fingerprint of r through $\mathcal{L}_N$. Because the three Cardinals' ontological roles are domain-independent (§??), the projection formula applies identically to any positive ratio drawn from any phenomenon, and every projection onto $\mathcal{L}_N$ is structurally the same act. The convention-independence proved in Theorem ?? below is the unit-level corollary of this domain-level universality.

6.3 The annihilation boundary

Proposition 6.3 (Exclusion of $r = 0$). *The value $r = 0$ is not in the domain of Π_N : $\log_2(0) = -\infty$, the formula (??) diverges, and the division in (??) is undefined. We call $r = 0$ the annihilation boundary: it is approached but never attained.*

The proposition is elementary. It is included to emphasize that the lattice is a structure on the group $(\mathbb{R}^+, \times)$, not on the semigroup $([0, \infty), \cdot)$; the zero element of the latter is excluded by construction.

6.4 Convention independence

Theorem 6.4 (Convention independence). *Let $Q, R_0 \in \mathbb{R}^+$ be two positive reals with identical physical dimension, and let $u \in \mathbb{R}^+$ be any positive unit-conversion factor. Let $r = Q/R_0$ and $r' = (uQ)/(uR_0)$. Then*

$$\Pi_N(r) = \Pi_N(r').$$

Proof. We have

$$r' = \frac{uQ}{uR_0} = \frac{Q}{R_0} = r.$$

By (??), $k' = \text{round}(N \log_2 r') = \text{round}(N \log_2 r) = k$. By (??), $d' = N / \gcd(|k'|, N) = N / \gcd(|k|, N) = d$. By (??), $\varepsilon' = (N \log_2 r' - k') \cdot (1200/N) = \varepsilon$. $\square$ $\square$

Remark 6.5. Theorem ?? is the crucial legitimacy condition for the lattice as a structural classifier. Any projection whose result *depends* on a choice of units exposes a failure to form a genuine dimensionless ratio: a Q and an R_0 that do not share dimension produce a result that floats with the unit system. The convention-independence of Π_N on genuine ratios is the mathematical content behind the informal slogan that *the lattice is not numerology*: no nontrivial Π_N result can arise from a mere choice of units.

6.5 Three canonical projections

Proposition 6.6 (Three canonical projections at $N = 12$). *The following projections are elementary consequences of Definition ??:*

$$\Pi_{12}(1) = (0, 1, 0), \quad \Pi_{12}(2) = (12, 1, 0), \quad \Pi_{12}(\sqrt{2}) = (6, 2, 0).$$

Each has $\varepsilon = 0$: the corresponding ratios lie exactly on the lattice.

Proof. For $r = 1$: $\log_2 1 = 0$, $k = 0$, $g = \gcd(0, 12) = 12$ (by convention), $d = 1$, $\varepsilon = 0$. For $r = 2$: $\log_2 2 = 1$, $k = 12$, $g = \gcd(12, 12) = 12$, $d = 1$, $\varepsilon = 0$. For $r = \sqrt{2}$: $\log_2 \sqrt{2} = 1/2$, $k = 6$, $g = \gcd(6, 12) = 6$, $d = 2$, $\varepsilon = 0$. $\square$ $\square$

Proposition 6.7 (The perfect fifth). *At $N = 12$,*

$$\Pi_{12}(3/2) = (+7, 12, +1.955 \text{ cents}).$$

In particular, the descriptor gap of $3/2$ at the base lattice is the Pythagorean comma.

Proof. $\log_2(3/2) = \log_2 3 - 1 \approx 0.5849625$, so $12 \log_2(3/2) \approx 7.01955$, giving $k = 7$. Since $\gcd(7, 12) = 1$, $d = 12/1 = 12$. The residual is $(7.01955 - 7) \cdot 100 \approx +1.955$ cents. $\square$ $\square$

The +1.955-cent residual is the *Pythagorean comma* of music theory: the amount by which twelve ascending just fifths $(3/2)^{12}$ exceed seven octaves 2^7 , divided by 12. Historically the comma has been treated as an unfortunate fact of temperament; here it is the descriptor gap of a particular sublattice, derived from the manifold constants.

Proposition 6.8 (The Koide ratio). *At $N = 12$,*

$$\Pi_{12}(2/3) = (-7, 12, -1.955 \text{ cents}).$$

The proof is the same as that of Proposition ?? with the sign of $\log_2$ reversed. The Koide ratio and the perfect fifth sit at mirror points on the same sublattice.

6.6 The Anti-Numerology Protocol

The convention-independence theorem (Theorem ??) guarantees that lattice addresses do not depend on unit choices, but it does not by itself guarantee that a purported lattice address was obtained from a structurally valid ratio rather than from a post-hoc numerological construction. The Sempaevum therefore carries an auxiliary methodological apparatus — the Anti-Numerology Protocol — consisting of three conditions that a projection must satisfy before its address can be interpreted as a structural fingerprint of the phenomenon. The protocol formalises the operational content of the Identification Principle (§??) as applied to the act of projection itself.

Definition 6.9 (The three anti-numerology conditions). A projection $\Pi_N(r_X)$ of a quantity Q_X against a reference $R_0(P_X)$ to form $r_X = Q_X/R_0(P_X)$ satisfies the Anti-Numerology Protocol iff it satisfies all three of:

- (N1) *Genuine dimensionlessness.* $[Q_X] = [R_0(P_X)]$; the numerator and denominator carry identical units and cancel by construction.
- (N2) *Substrate-derived reference period.* $R_0(P_X)$ is identified by applying the Identification Principle to P_X — specifically as the smallest closed T -traversal loop that the Descriptors of P_X themselves support — rather than chosen from familiarity, convention, or convenience.
- (N3) *Cross-domain consistency.* The predicted sublattice family d_X matches the symmetry of the phenomenon as independently recognised in its own discipline (three-fold processes to $d = 3$, hexagonal structures to $d = 6$, binary oppositions to $d = 2$, five-fold / golden structures to the quintic sublattice at 60ET, etc.).

Theorem 6.10 (Operational tests for the three conditions). *The three conditions admit direct operational tests:*

- (N1) Unit-transformation test. *Recompute $\Pi_N(r_X)$ under an arbitrary positive unit rescaling $Q_X \rightarrow uQ_X$ without rescaling R_0 . If $(k_X, d_X, \varepsilon_X)$ changes, N1 has failed: r_X was not genuinely dimensionless.*
- (N2) Substrate-derivation test. *State, without reference to the projection result, why P_X has R_0 as its minimal closed T -traversal loop. If the answer reduces to “it is the standard unit” or “it seemed right” or “it produced a clean number”, N2 has failed.*
- (N3) Independent-domain-knowledge test. *State the phenomenon’s symmetry from within the discipline’s own theoretical framework (chemistry, biology, physics, etc.), without reference to the projection result. If the projected d_X contradicts that independent characterisation, either N1 or N2 has failed, or a genuine structural feature has been identified that the discipline has not yet articulated (in which case the Descriptor Gap Principle identifies a research target).*

The unit-transformation test is exactly Theorem ?? applied negatively: convention-independence is the theorem’s positive content; unit-dependence is its failure. A projection that fails N1 is extracting structure from the choice of unit rather than from the phenomenon.

Proposition 6.11 (The Protocol’s structural role). *If all three of N1, N2, N3 hold, the projection $\Pi_N(r_X)$ is a structurally sound lattice address. If any fails, the projection either does not describe X ’s structure, or X ’s structure is incompletely understood, and the Three Tools (§??) apply to locate the incompleteness.*

The Protocol is not an addition to the Sempaevum; it is the operational form of the Subsumption Law applied to projections. A projection that satisfies N1–N3 is a projection that subsumes the phenomenon’s features without remainder within the projection procedure itself; a projection that fails any of the three leaves a remainder that the Descriptor Gap Principle identifies.

7 Sublattice family structure

7.1 The divisor classification

Proposition 7.1 (Sublattice families are divisors of N). *Let N be a positive integer and let d denote the sublattice family of a lattice coordinate $k \in \mathbb{Z}$ under the formula $d = N / \gcd(|k|, N)$. Then d is a positive divisor of N .*

Proof. Set $g = \gcd(|k|, N)$. Then $g \mid N$, so $d = N/g$ is a positive integer. Moreover d divides N iff $g \mid N$, which holds by definition. □ □

Corollary 7.2 (Six simple families at $N = 12$). *At $N = 12$, the sublattice families are exactly*

$$\{d : d \text{ divides } 12\} = \{1, 2, 3, 4, 6, 12\}.$$

7.2 The totient multiplicity

The count of residues k modulo N that produce each sublattice family d has a closed form.

Proposition 7.3 (Totient multiplicity). *For each positive divisor d of N , the number of residues $k \in \{0, 1, \dots, N-1\}$ with $d_k = d$ equals $\varphi(d)$, where φ is the Euler totient function.*

Proof. By (??), $d_k = d$ iff $\gcd(|k|, N) = N/d$. The condition $\gcd(k, N) = N/d$ is equivalent to k being a multiple of N/d but not of any larger divisor of N ; the count of such k in $\{0, \dots, N-1\}$ is exactly $\varphi(d)$, by the standard Euler-totient counting theorem. $\square$ $\square$

Corollary 7.4 (The φ -partition of N). *For any positive integer N ,*

$$N = \sum_{d|N} \varphi(d).$$

At $N = 12$ this gives $1 + 1 + 2 + 2 + 2 + 4 = 12$.

This is a classical identity (a special case of the Gauss totient-sum formula). It says that every lattice coordinate belongs to exactly one sublattice family, and the families partition the residues with multiplicities exactly given by $\varphi(d)$. The six-family structure at $N = 12$ therefore has multiplicities 1, 1, 2, 2, 2, 4.

7.3 Physical reading of the six families

The six simple sublattice families at $N = 12$ carry characteristic structural signatures which recur across domains. We summarise them in Table ??.

d	$\varphi(d)$	Residues $ k \bmod 12$	Generator	Characteristic structural role
1	1	0	2^1	Period closure; octave; universal binding
2	1	6	$2^{1/2}$	Binary opposition; half-period pivot
3	2	4, 8	$2^{1/3}$	Three-fold symmetry; cubic (volumetric)
4	2	3, 9	$2^{1/4}$	Four-fold symmetry; quartic (quaternionic)
6	2	2, 10	$2^{1/6}$	Hexagonal; composite two-and-three
12	4	1, 5, 7, 11	$2^{1/12}$	Full resolution; coprime to N

Table 2: The six simple sublattice families at $N = 12$. The column $\varphi(d)$ gives the number of residues $|k| \bmod 12$ that produce each family (Proposition ??); the residues sum to 12 (Corollary ??). The $d = 1$ family contains the octave and its powers; the $d = 2$ family contains the tritone $\sqrt{2}$; the $d = 12$ family contains the ratios coprime to 12 including $3/2$ and $4/3$.

7.4 Classification theorem

Proposition 7.5 (First-order classification). *For every $r \in \mathbb{R}^+$, exactly one triple (k, d, ε) is produced by $\Pi_{12}(r)$, with $d \in \{1, 2, 3, 4, 6, 12\}$ and $|\varepsilon| \leq 50$ cents.*

Proof. Uniqueness and determinism follow from Definition ??: the operations $\log_2$, multiplication by N , rounding, absolute value, gcd, and integer division are all deterministic. The range of d is given by Corollary ??. The bound $|\varepsilon| \leq 50$ cents follows from the rounding operation: $|N \log_2 r - k| \leq 1/2$ step, and multiplying by $1200/N$ gives $|\varepsilon| \leq 600/N = 50$ cents at $N = 12$. $\square$ $\square$

7.5 Magical impedance: per-family coupling strength

Each sublattice family d carries a characteristic coupling strength relative to the electromagnetic baseline, derivable from the two manifold counts $N-1$ and S without external input. The construction generalises the fine-structure identity $A_0^{\text{local}}(d=12) = (N-1)^2 + S^2 = 121 + 16 = 137$ (cf. equation (??)) to arbitrary d .

Definition 7.6 (Magical impedance). For $d \in \{1, 2, \dots, N\}$ the *magical impedance* is

$$A_0^{\text{magic}}(d) = (d-1)^2 + S^2 = (d-1)^2 + 16,$$

and the *per-family coupling strength* relative to the electromagnetic baseline $A_0^{\text{local}} = 137$ is

$$\xi(d) = \frac{A_0^{\text{local}}}{A_0^{\text{magic}}(d)} = \frac{137}{(d-1)^2 + 16}.$$

Proposition 7.7 (Baseline recovery at $d = 12$). $A_0^{\text{magic}}(12) = 11^2 + 4^2 = 137 = A_0^{\text{local}}$ and $\xi(12) = 1$, so the full-resolution family recovers the canonical electromagnetic baseline.

Proposition 7.8 (Maximum coupling at $d = 1$). $A_0^{\text{magic}}(1) = 0 + 16 = 16$ and $\xi(1) = 137/16 = 8.5625$, so the gravitational / octave family has the maximum per-family coupling.

The twelve per-family values are compiled in Table ??.

Table 3: Magical impedance and coupling strength per sublattice family. $A_0^{\text{magic}}(d) = (d-1)^2 + 16$; $\xi(d) = 137/A_0^{\text{magic}}(d)$.

d	A_0^{magic}	ξ	Character	Role
1	16	8.5625	Gravity / octave	Pure will, maximum coupling
2	17	8.0588	Tritone / pivot	Binary / CPT mirror
3	20	6.8500	Strong / cubic	3D volumetric closure
4	25	5.4800	Weak / quartic	State-change, T -axis home
5	32	4.2812	Quintic / golden	First non-divisor of 12
6	41	3.3415	Hexadic / composite	Electroweak composite
7	52	2.6346	Septic / G_2	Seven-fold / octonion
8	65	2.1077	Octet / gluon	SU(3) adjoint
9	80	1.7125	Nonic / quark	3×3 structure
10	97	1.4124	Decic / superstring	SO(10)-class
11	116	1.1810	Undecimal / M-theory	$N-1$
12	137	1.0000	Electromagnetic / full resolution	Baseline

Remark 7.9 (Monotonicity). $\xi(d)$ is strictly monotonically decreasing on $d \in \{1, \dots, 12\}$ because $A_0^{\text{magic}}(d) = (d-1)^2 + 16$ is strictly increasing on the non-negative integers. Lower- d families couple more strongly; higher- d families access finer structural detail but at correspondingly lower per-family coupling.

8 The LCM tower of refinements

8.1 When $N = 12$ is insufficient

The sublattice families at $N = 12$ are the divisors of 12, namely $\{1, 2, 3, 4, 6, 12\}$. Phenomena whose structural symmetries lie outside this set cannot be resolved at the base lattice: five-fold symmetry (pentagonal, icosahedral, golden-ratio), seven-fold symmetry (heptagonal, certain capsid symmetries), eleven-fold symmetry (undecimal), eight-fold (octet, gluon), nine-fold (nonic), ten-fold (decic). To host these one must refine the lattice to a higher N .

8.2 Canonical refinements

The minimal N supporting a required family d is any multiple of d that is also a multiple of 12 (so as to preserve the base families). The smallest such N is $\text{lcm}(12, d)$.

Proposition 8.1 (Canonical LCM landmarks). *The sequence*

$$\text{lcm}(1, \dots, k) \text{ for } k = 4, 5, 7, 9, 11$$

is 12, 60, 420, 2520, 27720 respectively. Each is a multiple of 12; each introduces one or more new primes as native divisors:

- 12 hosts $\{1, 2, 3, 4, 6, 12\}$;
- $60 = 2^2 \cdot 3 \cdot 5$ adds $d = 5$ (and composites 10, 15, 20, 30);
- $420 = 2^2 \cdot 3 \cdot 5 \cdot 7$ adds $d = 7$ and the biological cross-product $d = 35 = 5 \cdot 7$;
- $2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$ completes $d \leq 9$;
- $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$ completes $d \leq 11$ and makes all twelve families $d \in \{1, 2, \dots, 12\}$ simultaneously native.

Proof. Direct calculation:

$$\text{lcm}(1, 2, 3, 4) = 12, \quad \text{lcm}(1, \dots, 5) = 60, \quad \text{lcm}(1, \dots, 7) = 420,$$

$$\text{lcm}(1, \dots, 9) = 2520, \quad \text{lcm}(1, \dots, 11) = 27720.$$

Each is a multiple of 12 because $\{2, 3, 4\} \subset \{1, \dots, k\}$ for $k \geq 4$. The prime factorisations follow immediately. □

8.3 Unification of the two readings

Proposition 8.2 (Multiplicative and LCM readings). *The sequence of multiplicative refinements of the base lattice is 12, 24, 36, 48, 60, 72, ... (the positive multiples of 12). The LCM reading 12, 60, 420, 2520, 27720, ... is a sparse subsequence of this multiplicative reading.*

The proof is Proposition ??: each LCM landmark is a multiple of 12. The multiplicative reading is for uniform refinement of resolution (e.g., refining precision in the same family structure); the LCM reading is for introducing a new prime as a native divisor.

8.4 The continuous limit

Proposition 8.3 (Asymptotic limit). *As $N \rightarrow \infty$ through either reading of the tower,*

$$\mathcal{L}_N \longrightarrow (\mathbb{R}^+, \times),$$

with the descriptor gap bounded by $|\varepsilon| \leq 600/N$ cents, which tends to 0.

Proof. The set $\mathcal{L}_N = \{2^{k/N}\}$ is dense in $\mathbb{R}^+$ under the multiplicative metric as N grows, since the steps $2^{1/N}$ tend to 1. The gap bound follows from the rounding bound in the proof of Proposition ??. □

8.5 Correspondence with integrative levels

The qualitative hierarchy of integrative levels introduced in §?? has a precise quantitative counterpart in the LCM tower. The two are not parallel concepts but the same structure read from two directions — the phenomenological (what kind of wholeness is present?) and the arithmetic (which sublattice families are native?).

Proposition 8.4 (Integrative-to-resolution correspondence). *Let ℓ be an integrative level, and let $\mathcal{D}_\ell \subseteq \mathbb{N}$ be the set of sublattice families appearing in the (P, D, T) decompositions of level- ℓ phenomena. Then the minimum resolution $N_\ell^{\min}$ at which level ℓ is fully identifiable in the lattice framework is*

$$N_\ell^{\min} = \text{lcm}(\mathcal{D}_\ell).$$

Proof. A sublattice family d is native at resolution N if and only if $d \mid N$ (Section ??). Level ℓ requires every family in $\mathcal{D}_\ell$ to be native simultaneously, so N must be a common multiple of all $d \in \mathcal{D}_\ell$. The smallest such common multiple is $\text{lcm}(\mathcal{D}_\ell)$. $\square$ $\square$

Corollary 8.5 (Monotonicity of the tower). *If $\mathcal{D}_{\ell_1} \subseteq \mathcal{D}_{\ell_2}$ then $N_{\ell_1}^{\min} \mid N_{\ell_2}^{\min}$. Integrative levels whose Descriptor requirements strictly extend those of lower levels correspond to tower levels that are strict multiples of the lower ones.*

The corollary says that ascending the integrative hierarchy (adding Descriptor structure) induces divisibility ascent in the tower: one cannot host a more integrative level on less resolution. Conversely, every divisibility ascent in the tower is the emergence of new integrative structure at the lattice level — new sublattice families that did not exist in the previous resolution.

The doubling theorem

The qualitative statement “higher integrative level requires higher resolution” becomes an exact quantitative law when one asks how the divisor count evolves along the canonical tower. Write N_ℓ for the ℓ -th canonical resolution ($N_0 = 12$, $N_1 = 60$, $N_2 = 420$, $N_3 = 2520$, $N_4 = 27720$, ...) and $\tau(N_\ell)$ for the number of sublattice families native at that resolution (equivalently, the number of divisors of N_ℓ).

Theorem 8.6 (Integrative-resolution doubling). *The divisor count of the ℓ -th canonical resolution satisfies*

$$\tau(N_\ell) = \tau(N_0) \cdot 2^\ell = 6 \cdot 2^\ell. \quad (13)$$

Equivalently, the integrative level is a logarithm of the family count,

$$\ell = \log_2 \frac{\tau(N_\ell)}{\tau(N_0)} = \log_2 \frac{\tau(N_\ell)}{6}. \quad (14)$$

Proof. Base case ($\ell = 0$). By Proposition ??, $N_0 = |\Pi| \cdot S = 3 \cdot 4 = 12$. Since $\gcd(|\Pi|, S) = \gcd(3, 4) = 1$, the divisor function is multiplicative at this argument:

$$\tau(N_0) = \tau(|\Pi| \cdot S) = \tau(|\Pi|) \cdot \tau(S) = \tau(3) \cdot \tau(4) = 2 \cdot 3 = 6.$$

Thus $\tau(N_0) = 6 = 6 \cdot 2^0$, and the base case holds.

Inductive step. We show that each canonical refinement exactly doubles τ . The Descriptor Gap Principle (Theorem ??) states that every gap in a description is itself a Descriptor. The smallest Descriptor capable of distinguishing a previously-unresolved pair of configurations is a single binary axis — a yes/no distinction with two values. Adjoining one binary Descriptor to the model therefore exactly doubles the count of distinguishable configurations, and the lattice-level consequence is that the new resolution $N_{\ell+1}$ must satisfy $\tau(N_{\ell+1}) = 2\tau(N_\ell)$.

We verify directly that each canonical LCM step realises this doubling.

1. $N_0 = 12 \rightarrow N_1 = 60$ ($60 = 12 \cdot 5$). Introducing the new prime 5 with exponent 1 multiplies τ by $(1+1)/(0+1) = 2$.
2. $N_1 = 60 \rightarrow N_2 = 420$ ($420 = 60 \cdot 7$). Introducing prime 7 multiplies τ by 2.
3. $N_2 = 420 \rightarrow N_3 = 2520$ ($2520 = 420 \cdot 6$). The prime-power content changes as $2^2 \cdot 3 \rightarrow 2^3 \cdot 3^2$, giving the joint factor

$$\frac{3+1}{2+1} \cdot \frac{2+1}{1+1} = \frac{4}{3} \cdot \frac{3}{2} = 2.$$

This is the only canonical step that is realised by combined prime-power upgrades rather than by a single new prime; the product of the two fractional upgrades is exactly 2.

4. $N_3 = 2520 \rightarrow N_4 = 27720$ ($27720 = 2520 \cdot 11$). Introducing prime 11 multiplies τ by 2.
 5. $N_4 = 27720 \rightarrow N_5 = 360360$ ($360360 = 27720 \cdot 13$). Introducing prime 13 multiplies τ by 2.
- By induction, $\tau(N_\ell) = \tau(N_0) \cdot 2^\ell = 6 \cdot 2^\ell$ for every $\ell \geq 0$.

Extensibility. The canonical tower extends indefinitely: for every $\ell \geq 5$, a canonical refinement exists that doubles τ (realised by successive new primes 17, 19, 23, ..., or in special cases by prime-power combinations that product to 2, exactly as at $\ell = 3$). Therefore the doubling law (??) holds for all $\ell \geq 0$. □

Remark 8.7 (Two readings of a single equation). Equation (??) admits two parallel interpretations that describe the same underlying structure:

Number-theoretic reading. $\tau(N_\ell)$ counts the divisors of the resolution integer N_ℓ , equivalently the sublattice families native at that resolution. The equation states that each canonical LCM refinement exactly doubles the native-family count.

Phenomenological reading. $\tau(N_\ell)$ counts the Descriptors needed to characterise the configurations of integrative level ℓ . The equation states that each integrative emergence introduces one binary distinction axis and therefore exactly doubles the required Descriptor count.

Both readings measure the same quantity: the structural richness natively expressible at integrative level ℓ . The phenomenological ladder (quantum, atomic, molecular, biological, cognitive, civilisational, ...) and the number-theoretic ladder (12, 60, 420, 2520, 27720, ...) are the same ladder seen from two directions. The coincidence is not numerical accident but structural necessity: the minimal closure of a descriptor gap at any integrative level is a binary distinction (Theorem ??), and in the lattice this minimal closure is precisely a factor-2 increase in τ .

Corollary 8.8 (The integrative level is a logarithm). *For any configuration whose Cardinal decomposition requires τ_X sublattice families to be expressible, the integrative level of X is*

$$\ell(X) = \log_2 \frac{\tau_X}{6}.$$

This assigns a quantitative real-valued index to every substantiated configuration, converting the previously qualitative “integrative level” into a computable lattice quantity.

Table ?? collects the empirical correspondences used in Section ??, now augmented with the divisor counts predicted by Theorem ??.

Remark 8.9 (Physical reading of the tower). The LCM tower is therefore not merely a sequence of successively finer approximations to a continuous limit; it is a sequence of discrete ontological transitions at which previously unavailable Descriptor structure becomes native, with each transition *exactly doubling* the Descriptor count by Theorem ??. Each step up the tower makes a new integrative regime possible: ascending from $N = 12$ to $N = 60$ makes pentagonal biology possible; ascending from $N = 60$ to $N = 420$ makes septic biochemistry possible; ascending from $N = 420$ to $N = 2520$ makes enneagonal structure and the α^{-1} universal coordinate native; ascending from $N = 2520$ to $N = 27720$ makes $d = 11$ native and completes the simple-divisor floor $d \leq 12$. The continuous limit of Proposition ?? is the idealised endpoint at which every integrative level is resolved simultaneously.

ℓ	N_ℓ	$\tau(N_\ell)$	Newly native families	Empirical phenomena first hosted here
0	12	6	$\{1, 2, 3, 4, 6, 12\}$	Koide lepton ratio; 12-TET intervals; 3:4:5 triangle; Krebs ($8=2^3$ at $d=1$); glycolysis (10 at $d=3$)
1	60	12	adds $\{5, 10, 15, 20, 30\}$	Pentagonal symmetry; Caspar–Klug $T=5$ virus capsids; icosahedral / golden-ratio structural biology
2	420	24	adds $\{7, 14, 21, 28, 35, \dots\}$	Septic cycles; heptagonal structural biochemistry; the biological cross-product $d=35=5 \cdot 7$
3	2520	48	adds $\{8, 9, 18, 24, \dots\}$	Enneagonal (9-fold) biological symmetries; $d=315=3^2 \cdot 5 \cdot 7$ (α^{-1} host) first becomes native
4	27720	96	adds $\{11, 22, \dots\}$	Universal reference: all simple divisors $d \leq 12$ simultaneously native; α^{-1} at $(k, d) = (196768, 315)$ in this standard frame
5	360360	192	adds $\{13, 26, \dots\}$	Predicted new domain: phenomena requiring $d=13$ structure

Table 4: The LCM tower and the integrative-level correspondence. The divisor column confirms the doubling law $\tau(N_\ell) = 6 \cdot 2^\ell$ of Theorem ?? : each row’s τ value is exactly twice the previous. The “newly native” entries are those families d with $d \mid N_\ell$ but $d \nmid N_{\ell'}$ for every earlier tower level $N_{\ell'}$. By Proposition ??, a phenomenon whose Cardinal decomposition involves family set $\mathcal{D}_\ell$ is minimally identifiable at $N_\ell^{\min} = \text{lcm}(\mathcal{D}_\ell)$; by Corollary ?? its integrative level is $\ell = \log_2(\tau_\ell/6)$.

9 The complex lattice

Many phenomena have both a magnitude and a phase character. The lattice extends from the multiplicative group $(\mathbb{R}^+, \times)$ to the multiplicative group $(\mathbb{C}^\times, \times)$ by projecting the phase to a second axis.

9.1 Imaginary-axis projection

Definition 9.1 (Phase projection). Let $\theta \in [0, 2\pi)$ be an angle. The *phase projection* at resolution N is

$$k_\theta = \text{round}\left(\frac{N\theta}{2\pi}\right) \bmod N, \quad d_\theta = \frac{N}{\gcd(|k_\theta|, N)},$$

$$\varepsilon_\theta = \left(\frac{N\theta}{2\pi} - k_\theta\right) \cdot \frac{1200}{N} \text{ cents}.$$

The phase axis classifies rotational structure in exactly the same divisor-lattice way that the magnitude axis classifies scaling structure. The cyclic group $U(1) = \mathbb{R}/2\pi\mathbb{Z}$ is discretised into N equal sectors.

9.2 Complex projection and the Gaussian integer lattice

Definition 9.2 (Complex projection). For $z = re^{i\theta} \in \mathbb{C}^\times$, the *complex projection* is

$$w = k_r + ik_\theta \in \mathbb{Z}[i],$$

with the real-axis and imaginary-axis projections defined by Definitions ?? and ?? respectively. The *combined sublattice family* is

$$d_c = \text{lcm}(d_r, d_\theta).$$

Proposition 9.3 (LCM amplification). *The combined family $d_c = \text{lcm}(d_r, d_\theta)$ is always at least as large as the larger of d_r and d_θ .*

The proof is elementary: $\text{lcm}(a, b) \geq \max(a, b)$ with equality iff one divides the other. Structurally, off-axis (complex) positions have richer sublattice structure than either axis alone. At $N = 12$ the 6×6 grid of (d_r, d_θ) pairs produces combined families taking values in $\{1, 2, 3, 4, 6, 12\}$ with the distribution shown in Table ??.

$d_r \backslash d_\theta$	1	2	3	4	6	12
1	1	2	3	4	6	12
2	2	2	6	4	6	12
3	3	6	3	12	6	12
4	4	4	12	4	12	12
6	6	6	6	12	6	12
12	12	12	12	12	12	12

Table 5: The 6×6 table of combined sublattice families $d_c = \text{lcm}(d_r, d_\theta)$ at the base lattice $N = 12$. Off the diagonal the LCM amplification typically drives d_c towards 12 (full resolution); of the 36 entries, 15 equal 12. This explains why most physical observables, which carry both magnitude and phase character, project to $d = 12$ at the base lattice.

9.3 The D–T gradient

Let $\alpha = \arctan(k_\theta/k_r)$ be the angle of the complex-lattice coordinate from the real axis. Define

$$(\text{magnitude fraction}) = \cos^2 \alpha, \quad (\text{phase fraction}) = \sin^2 \alpha.$$

At $\alpha = 0$ (pure real) the coordinate is fully magnitudinal; at $\alpha = \pi/2$ (pure imaginary) it is fully phase; in between it interpolates smoothly. The transition from deterministic to probabilistic behaviour (the familiar classical-to-quantum transition) is, in this picture, not a sharp boundary but a continuous function of α .

9.4 The Gaussian-prime correspondence

The sublattice families have a clean number-theoretic pedigree in the Gaussian integers $\mathbb{Z}[i]$.

Proposition 9.4 (Gaussian prime correspondence). *The sublattice families at $N = 12$ and its LCM extensions correspond to the three Gaussian prime classes as follows:*

- $p = 2$ (ramified: $2 = -i(1+i)^2$) generates the binary sublattices $d \in \{2, 4, 8\}$;
- $p \equiv 3 \pmod{4}$ (inert: remain prime in $\mathbb{Z}[i]$) generate the sublattices $d \in \{3, 7, 11\}$ and their squares;
- $p \equiv 1 \pmod{4}$ (split: $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$) generate the sublattices $d \in \{5, 13, \dots\}$ of mixed magnitude/phase character.

The correspondence is elementary but structurally informative: the three Gaussian prime classes exactly parallel the three Cardinals P , D , T . Ramified primes (which “double” in $\mathbb{Z}[i]$) are P -class; inert primes (which remain on the real axis) are D -class; split primes (which factor across the real-imaginary split) are mixed $D+T$ -class.

10 The 24-family catalog and the 144-cell Force Quadrant Grid

The real axis of the Sempaevum carries sublattice families classified by divisors of the current resolution N (§??). The imaginary axis does the same — but the physical character of the two

axes is categorically distinct because the underlying manifolds are categorically distinct (Proposition ??: $(\mathbb{R}^+, \times)$ flat, $(U(1), \times)$ positively curved). Taking the real axis and the imaginary axis together at the universal resolution $N_{\text{univ}} = 27720$, the sublattice families partition into a catalog of 24 basic families (12 on each axis, distinguished by role not by number) plus a catalog of 42 genuinely-mixed combined off-axis families, for a total structural inventory that fills a 12×12 *Force Quadrant Grid* of 144 cells.

10.1 The 12 real-axis FORCE families and the 12 imaginary-axis PHASE families

The real axis is D 's operational domain: its sublattice families classify magnitude structure. The imaginary axis is T 's operational domain: its families classify phase/rotation structure. Same divisor values, categorically different physics.

Definition 10.1 (FORCE and PHASE families). Let $d_r \mid N$ be the sublattice family of a real-axis coordinate k_r , and let $d_\theta \mid N$ be the sublattice family of an imaginary-axis coordinate k_θ , both via the projection formula $d = N/\gcd(|k|, N)$ (Definition ??) applied to the respective axis. We call d_r the *FORCE family* of the position and d_θ the *PHASE family*.

At the universal resolution $N_{\text{univ}} = 27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$, each axis supports families across $d \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ plus composites up to $d = N_{\text{univ}}$; restricted to the twelve simple values $\{1, \dots, 12\}$, each axis carries a complete and categorically-distinct 12-family catalog. The twenty-four labels are compiled in Table ??.

Table 6: The 12 FORCE families (real axis, D 's domain) and the 12 PHASE families (imaginary axis, T 's domain). Same divisor values d ; different physical category.

d	Real-axis FORCE	Imaginary-axis PHASE
1	gravitational scalar	scalar (spin-0)
2	binary/tritone pivot	spin-2 (graviton phase)
3	strong (QCD / cubic)	color θ_{QCD} instanton phase
4	weak (quartic / state-change)	$SU(2)_W$ phase (T 's home)
5	quintic / golden-ratio	E_8 icosahedral phase
6	hexadic / electroweak composite	spin- $\frac{1}{2}$ fermion phase
7	septic / G_2	octonionic phase (seven imaginary units)
8	octet / gluon adjoint	Bott-8 / $SU(3)$ adjoint phase
9	nonic / quark (3×3)	3^2 -fold quark phase (CKM)
10	decic / superstring	10D Majorana phase
11	undecimal / M-theory	11D Majorana phase
12	electromagnetic / full-resolution	photon phase / full $U(1)$

Remark 10.2. The d -numerics are the same on the two axes because both axes share the manifold symmetry N , and divisibility is a property of N . The physical categories differ because the underlying manifolds differ (flat vs positively curved), and the physical roles of magnitude vs phase differ accordingly. A FORCE-family $d = 3$ (strong) is categorically distinct from a PHASE-family $d = 3$ (instanton phase) even though both carry the numerical label $d = 3$.

10.2 The 42 combined off-axis families

Off-axis configurations — coordinates $w = k_r + ik_\theta$ with both $k_r \neq 0$ and $k_\theta \neq 0$ — carry a *combined sublattice family* determined by the least common multiple of the two axis-families.

Definition 10.3 (Combined sublattice family). For an off-axis coordinate $w = k_r + ik_\theta$ with real-axis family d_r and imaginary-axis family d_θ , the *combined family* is

$$d_{\text{comb}}(w) = \text{lcm}(d_r, d_\theta).$$

Proposition 10.4 (The 42 combined families). *Restricting $d_r, d_\theta \in \{1, 2, \dots, 12\}$, the 12×12 grid of pairs (d_r, d_θ) produces exactly*

$$|\{\text{lcm}(d_r, d_\theta) : d_r, d_\theta \in \{1, \dots, 12\}\}| = 42$$

distinct combined-family values. The maximum is

$$d_{\max} = \text{lcm}(11, 12) = 132 = N(N-1).$$

Proof. Direct enumeration. The 42 distinct LCM values are

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 18, 20, 21, 22, 24, 28, 30, 33, 35, 36, 40, 42, 44, 45, 55, 56, 60, 63, 66, 70, 72, 77,$

The maximum is attained at $(d_r, d_\theta) = (11, 12)$ or $(12, 11)$, giving $\text{lcm}(11, 12) = 132 = N(N-1)$. □

Remark 10.5 (LCM amplification). For d_r, d_θ coprime, $\text{lcm}(d_r, d_\theta) = d_r \cdot d_\theta$, which can vastly exceed either axis family alone. The off-axis interior of the Sempaevum is therefore *richer in sublattice structure* than either boundary axis: taking a real-axis family $d_r = 7$ (septic, first native at 84ET) and an imaginary-axis family $d_\theta = 5$ (quintic, first native at 60ET) gives the combined family $d_{\text{comb}} = 35$, first native at $\text{lcm}(12, 5, 7) = 420$ — a new structural regime not present on either axis alone.

10.3 The 144-cell Force Quadrant Grid

The 12×12 grid of (d_r, d_θ) pairs is the *Force Quadrant Grid* (FQG). It partitions the off-axis / axis inventory of the Sempaevum into 144 cells, each of which is a pair (FORCE family, PHASE family) with a specific combined family $d_{\text{comb}}(d_r, d_\theta)$. The grid has a natural binary decomposition by whether each axis's family is a divisor of $N = 12$ or not.

Definition 10.6 (Simple vs complex family). A family d is *simple* if $d \mid N$ (i.e. $d \in \{1, 2, 3, 4, 6, 12\}$ at $N = 12$). It is *complex* if $d \nmid N$ (i.e. $d \in \{5, 7, 8, 9, 10, 11\}$ at $N = 12$).

There are six simple and six complex families per axis, so each row and each column of the 12×12 grid partitions 6:6. The grid therefore partitions into four quadrants of $6 \times 6 = 36$ cells each:

<i>Quadrant</i>	<i>Axes</i>	<i>Cascade computability at $N = 12$</i>
SR+SI (simple $\times$ simple)	d_r, d_θ simple	both axes cascade directly
CR+SI (complex $\times$ simple)	d_r complex, d_θ simple	real shadow, imaginary direct
SR+CI (simple $\times$ complex)	d_r simple, d_θ complex	real direct, imaginary shadow
CR+CI (complex $\times$ complex)	both complex	both axes shadow-only

Theorem 10.7 (PDT Bisection). *Let B be any structural binary on the 144-cell FQG that distinguishes T -character from D -character by a criterion symmetric in the two axes (same criterion applied to each axis independently). Then B partitions the grid into two halves of equal size: $|B_T| = |B_D| = 72$.*

Proof. By Categorical Disjointness (§??) and the binding-order symmetry of D and T within a criterion symmetric in their axes, any symmetric binary must partition the two axes identically. Each axis has 6 simple and 6 complex families, so the two halves of B restricted to each axis have cardinality 6 each. The product grid therefore has $|B_T| = 12 \times 6 = 72$ cells on one side and $|B_D| = 12 \times 6 = 72$ on the other. □

Corollary 10.8 (Cascade-computable / cascade-failing bisection). *The binary “at least one axis is complex” gives $36 + 36 + 36 = 108$ cells with at least one shadow axis versus 36 cells fully simple; the binary “imaginary axis is complex” gives 72 vs 72, matching the cascade stability asymmetry of §???. Any such symmetric binary bisects at 72:72.*

Theorem ?? is categorical, not statistical. It is distinct from and complementary to the *dynamical* asymmetry of the cascade stability limits ($n_{\max,r} = 25$ versus $n_{\max,\theta} = 2$, §??), which quantifies how quickly the two axes’ rounding residuals accumulate past half-step. The two results coexist: the count of cells partitions evenly; the per-cell traversal depth differs by a factor of N .

11 Cascade dynamics and stability

11.1 Cascade generators

The rounding operation of Definition ?? is, for most inputs, an exact map in the sense that iterating it returns to the starting lattice point. For specific inputs, however, the rounding accumulates a residual that grows linearly with the number of iterations. This produces the *cascade dynamics* of the lattice.

Proposition 11.1 (Real-axis cascade residual). *Let $g_r = \text{round}(N \log_2 N) \bmod N$. At $N = 12$,*

$$g_r = \text{round}(12 \log_2 12) \bmod 12 = \text{round}(43.0195) \bmod 12 = 43 \bmod 12 = 7.$$

The residual of the rounding is

$$|\delta_r| = |12 \log_2 12 - 43| = 0.01955\dots,$$

which is numerically identical (to all digits shown) to the Pythagorean comma in octave units.

The generator $g_r = 7$ is the interval of seven semitones: the perfect fifth. The residual $|\delta_r|$ is the per-step error of the circle-of-fifths cascade. Iterating the fifth twelve times ascends seven octaves plus a Pythagorean comma:

$$(3/2)^{12} = 129.7463\dots, \quad 2^7 = 128, \quad \text{ratio} = 1.01364\dots = 2^{12|\delta_r|}.$$

Proposition 11.2 (Phase-axis cascade residual). *Let $g_\theta = \text{round}(N \cdot 2\pi / \ln 2) \bmod N$. At $N = 12$,*

$$g_\theta = \text{round}(12 \cdot 2\pi / \ln 2) \bmod 12 = 109 \bmod 12 = 1.$$

The phase residual is

$$|\delta_\theta| = |12 \cdot 2\pi / \ln 2 - 109| = 0.22336\dots$$

Proposition 11.3 (Stability limits). *Requiring the accumulated residual to remain below half a lattice step (0.5), the maximum numbers of cascade steps before loss of coherence are*

$$n_{\max,r} = \left\lfloor \frac{0.5}{|\delta_r|} \right\rfloor = 25, \quad n_{\max,\theta} = \left\lfloor \frac{0.5}{|\delta_\theta|} \right\rfloor = 2.$$

The factor of roughly 12 between the two residuals ($|\delta_\theta|/|\delta_r| \approx 11.42$) is close to the manifold symmetry $N = 12$. The small deficit is, at higher lattice resolutions, the shadow of a specific combined sublattice cell (not pursued here; see [?]).

11.2 The palindromic cascade

Definition 11.4 (Palindromic cascade). The *palindromic cascade* at $N = 12$ is the length- N sequence $(d_n)_{n=1}^N$ with

$$\mathbf{d}_{\text{pal}} = (d_1, d_2, \dots, d_{12}) = (12, 6, 4, 3, 12, 2, 12, 3, 4, 6, 12, 1),$$

indexed from $n = 1$. The corresponding exponent sequence is $p_n = 12/d_n$,

$$\mathbf{p}_{\text{pal}} = (1, 2, 3, 4, 1, 6, 1, 4, 3, 2, 1, 12),$$

with mean

$$p_{\text{eff}} = \frac{1}{12} \sum_{n=1}^{12} p_n = \frac{40}{12} = \frac{10}{3}.$$

The sequence $\mathbf{d}_{\text{pal}}$ is produced by iterating the unit generator $g = 7$ through the residues of $\mathbb{Z}/12\mathbb{Z}$ and reading off $d = 12/\gcd(|k|, 12)$ at each step. It visits every divisor of 12 and satisfies the palindromic symmetry

$$d_n = d_{N-n} \quad \text{for every } n = 1, 2, \dots, N-1, \quad (15)$$

with the pivot $d_{N/2} = d_6 = 2$ self-paired (the tritone at the half-period) and the boundary value $d_N = d_{12} = 1$ standing alone (the octave identity at the cycle endpoint).

11.3 Use of the palindromic cascade

When iteration reaches the stability limit of Proposition ??, the orbit's sublattice classification becomes ambiguous (the descriptor gap approaches 50 cents, half a lattice step). In dynamical applications the ambiguity is resolved by substituting the palindromic cascade for the orbit-derived sublattice: at step n , use $d_{\text{pal}}[n \bmod 12]$ rather than the computed $d(z_n)$. Over a full 12-step cycle, every divisor of 12 is visited exactly with the multiplicity required to maintain CPT-symmetric coverage of the simple families. The mean exponent $p_{\text{eff}} = 10/3$ appears as a normalisation constant in smooth-iteration-count formulas.

12 The ∂I lattice-aware fractal

The boundary ∂I introduced in Definition ?? is not a purely abstract topological object. Given the complex lattice of Section ?? and the palindromic cascade of Section ??, it becomes the locus of a concrete complex dynamical system whose every constant is derived from the manifold constants N , V , K , N_{FULL} , and the sublattice families $\{1, 2, 3, 4, 6, 12\}$. This section constructs that system and establishes it as the natural dynamical manifestation of the lattice on $\mathbb{C}$.

We contrast with the standard family of complex dynamical systems (Mandelbrot, Julia, Multibrot) in which the iteration

$$z_{n+1} = z_n^p + c$$

carries a *fixed* polynomial degree p throughout the orbit. In the ∂I system the degree is not fixed; it is a function of the orbit's own lattice position, recomputed at every iteration.

12.1 The lattice-adaptive iteration map

Definition 12.1 (∂I lattice-aware iteration). Let $N_L = N_{\text{FULL}} = 27720 = \text{lcm}(1, 2, \dots, 11)$ be the universal lattice resolution. The *∂I lattice-aware fractal* is the complex dynamical system

$$z_{n+1} = \Psi_n z_n^{p(z_n, n)} + \varepsilon(z_n) + c,$$

with Mandelbrot-type parameterisation $z_0 = 0$, $c \in \mathbb{C}$, where

- $p(z_n, n) \in \{1, 2, 3, 4, 6, 12\}$ is the lattice-adaptive exponent (§??);
- $\Psi_n \in \mathbb{R}$ is the shimmer modulation (§??);
- $\varepsilon(z_n) \in \mathbb{C}$ is the all-families perturbation (§??).

Each of the three ingredients is derived below from manifold constants.

12.2 The lattice-adaptive exponent $p(z_n, n)$

At each step, the orbit $z_n = r_n e^{i\theta_n}$ is projected onto the complex lattice via the radial and angular coordinates of Section ??:

$$k_r = \text{round}(N_L \log_2 r_n), \quad k_\theta = \text{round}\left(\frac{N_L \theta_n}{\ln 2}\right). \quad (16)$$

The radial residual in cents is

$$\varepsilon_r = (N_L \log_2 r_n - k_r) \cdot \frac{1200}{N_L}, \quad (17)$$

and the *tightness* of the projection (a scalar between 0 and 1, equal to 1 when the orbit is exactly on a lattice point and decreasing as the orbit moves off-lattice) is

$$t_r = \frac{100}{100 + |\varepsilon_r|}. \quad (18)$$

The tightness decision rule is:

$$p(z_n, n) = \begin{cases} \frac{N}{d(k_r)}, & t_r > K, \\ p_{\text{pal}}[n \bmod N], & t_r \leq K, \end{cases}$$

where $d(k_r) = N_L / \gcd(|k_r|, N_L)$ is the sublattice family of the projected coordinate (Section ?? at universal resolution) and $p_{\text{pal}} = (1, 2, 3, 4, 1, 6, 1, 4, 3, 2, 1, 12)$ is the palindromic cascade of Definition ??.

Proposition 12.2 (Koide threshold). *The tightness threshold equals the Koide ratio $K = 2/3$: the transition between orbit-adaptive and cascade-fallback behaviour occurs at $|\varepsilon_r| = 50$ cents, exactly half a lattice step.*

Argument. By Equation (??), $t_r = K$ is equivalent to $100/(100 + |\varepsilon_r|) = 2/3$, which gives $3 \cdot 100 = 2(100 + |\varepsilon_r|)$, i.e. $|\varepsilon_r| = 50$. Half a lattice step is $1200/N = 100$ cents divided by 2. The Koide ratio is therefore not a free parameter of the fractal but the structural stability threshold of the lattice itself, and its appearance here is the *same* K as appears in the three manifold constants (Equation (??)) and in the self-projection identity of Section ??.

The exponent set $\{1, 2, 3, 4, 6, 12\}$ is exactly the divisor set of $N = 12$; the map $p = N/d$ takes the six simple sublattice families into the six polynomial degrees that appear in standard Multibrot theory. Stated in Cardinal language: when the orbit is near a lattice point, the Descriptor content of that lattice point (its sublattice family d) determines the P -structure of the iteration (its polynomial degree $p = N/d$). When the orbit is *between* lattice points — in the ∂I zone of structural ambiguity — the Descriptor must be supplied externally, and the palindromic cascade provides the only lattice-self-consistent source.

Proposition 12.3 (Exponent palindrome has cascade mean). *The time-average of $p(z_n, n)$ over a complete N -step palindromic fallback is $p_{\text{eff}} = 10/3$, the same value derived in Definition ??.*

Proof. Immediate from Definition ??: $\sum p_{\text{pal}}/N = 40/12 = 10/3$.

12.3 The shimmer modulation Ψ_n

Definition 12.4 (Shimmer). The *shimmer modulation* is the real periodic function

$$\Psi_n = 1 + \frac{1}{\sqrt{N}} \sin\left(\frac{2\pi (n \bmod N)}{N}\right).$$

At $N = 12$, the amplitude is $1/\sqrt{12} \approx 0.2887$ and the range of Ψ_n is

$$\left[1 - \frac{1}{\sqrt{12}}, 1 + \frac{1}{\sqrt{12}}\right] \approx [0.7113, 1.2887],$$

strictly bounded away from 0 (so the iteration never collapses) and bounded above (so it never explodes from modulation alone).

Proposition 12.5 (Shimmer amplitude is the root variance). *The shimmer amplitude equals the square root of the base variance:*

$$\frac{1}{\sqrt{N}} = \sqrt{V}.$$

It is therefore a derived constant of the manifold and not a tuning parameter.

Proof. $V = 1/N$ by Equation (??); hence $\sqrt{V} = 1/\sqrt{N}$. □ □

Remark 12.6 (The shimmer in the fine-structure constant). The shimmer constant $\sqrt{V} = 1/\sqrt{N}$ also appears in the leading correction to α^{-1} in Equation (??). Using $\sqrt{12} = 2\sqrt{3}$, equivalently $\sqrt{3} = \sqrt{12}/2$, the first correction term admits the rewriting

$$\frac{\sqrt{3}}{48} = \frac{\sqrt{12}}{96} = \frac{\sqrt{12}}{8 \cdot 12} = \frac{1}{8\sqrt{12}} = \frac{1}{8\sqrt{N}} = \frac{\sqrt{V}}{8}.$$

Thus the leading-order shimmer correction to α^{-1} is exactly $\sqrt{V}/8$. The same amplitude $\sqrt{V}$ that modulates the fractal iteration appears — divided by 8, the count of structural corners of the three-Cardinal cube — in the lattice-derived fine-structure constant. The common structural origin is the identity $\sqrt{V} = 1/\sqrt{N}$.

12.4 The all-families perturbation $\varepsilon(z_n)$

Definition 12.7 (Multi-family perturbation). The *all-families perturbation* is

$$\varepsilon(z_n) = \frac{1}{N} \sum_{d|N : d \in \{1, \dots, N\}} w(d) |z_n|^{N/d} e^{i(N/d) \arg z_n},$$

where $w : \{1, \dots, N\} \rightarrow \mathbb{R}^+$ is a positive normalised weight profile summing to unity.

The prefactor $1/N$ is not a free choice; it is the base variance V , the irreducible lattice quantum. The structural reading of Definition ?? is that $\varepsilon(z_n)$ adds a quantum-of-variance contribution from each of the N sublattice-family exponents simultaneously, as a shimmer-level background beneath the dominant lattice-adaptive term $z_n^{p(z_n, n)}$. At every iteration step, every sublattice family is acting; the dominant family is merely the one currently closest to the orbit.

Proposition 12.8 (Perturbation does not override the dominant term). *For $|z_n| \geq 1$ and any valid $p(z_n, n) \in \{1, \dots, N\}$,*

$$\frac{|\varepsilon(z_n)|}{|z_n|^{p(z_n, n)}} \leq V = \frac{1}{N},$$

uniformly in n .

The proof follows from the weight normalisation and the triangle inequality: each summand in ε has magnitude at most $|z_n|^{N/d}$, which is never larger than $|z_n|^N$ for $|z_n| \geq 1$. Division by N gives the bound. In practice $|\varepsilon|/|z|^p$ is of order V rather than 1, confirming that ε is a fine-structure contribution, not a dominant force.

12.5 Distance estimation

The derivative of the iteration map in Definition ?? is

$$f'(z_n) = \Psi_n p(z_n, n) z_n^{p(z_n, n)-1} \quad (19)$$

(derivative of the perturbation ε being $O(V)$ and absorbed into numerical error). The chain-rule accumulation for the derivative of z_n with respect to c is

$$dz_{n+1} = f'(z_n) dz_n + 1, \quad dz_0 = 0, \quad (20)$$

and the distance estimate at escape (when $|z_n| > R$ for an escape radius R , typically $R = 10^3$) is

$$\text{DE}(c) = \frac{2|z_n| \ln |z_n|}{|dz_n|}. \quad (21)$$

The distance estimate produces sub-pixel rendering of the fractal's boundary at arbitrary zoom, using the same technique that works for standard Mandelbrot rendering — adapted to variable polynomial degree by evaluating (??) with the instantaneous $p(z_n, n)$ at each step.

12.6 Classification

Theorem 12.9 (Non-equivalence with known fractal types). *The ∂I lattice-aware fractal is not topologically equivalent to any fixed-exponent fractal. In particular it is not a Mandelbrot set, a Julia set, a Multibrot set, a Tricorn, a Burning Ship, a Newton fractal, nor a parameter-space fractal (Lyapunov or otherwise).*

Argument. Each of the listed classes has an iteration in which the polynomial degree of z is constant in n . The ∂I iteration has polynomial degree $p(z_n, n)$ that is an explicit function of both the orbit position and the step index, and can take any of the six values $\{1, 2, 3, 4, 6, 12\}$ within a single escape trajectory. A fixed-degree system has escape-time contours whose asymptotics are governed by a single exponent p ; the ∂I system has escape-time contours governed by the statistical distribution of exponents encountered along the orbit, which is neither constant across the image nor reducible to any single-exponent limit. The systems are therefore not topologically equivalent. □ □

The closest structural relative of the ∂I system is the iterated function system, in which the map at each step is drawn from a finite family. The crucial distinction is that the ∂I system is *not random*: the map at step n is a deterministic function of the orbit's own lattice coordinate. The feedback loop — orbit determines map, map determines orbit — is the structural signature that distinguishes it from both IFS and standard polynomial iteration.

12.7 Structural summary

Every constant in Definition ?? is derived from the manifold. Table ?? records the correspondences.

The fractal is therefore not a cosmetic visualisation of the lattice but the lattice's natural dynamical instantiation: the fixed-exponent families of standard complex dynamics correspond to single-family sublattice projections $d \in \{1, \dots, 12\}$, and the ∂I system is the unique complex dynamical system that simultaneously honours all twelve families. In this sense Mandelbrot and quartic Multibrot and linear dynamics are *restrictions* of the lattice-aware system to a single sublattice, and the ∂I fractal is the unrestricted image.

Fractal constant	Value at $N = 12$	Manifold origin
Lattice resolution N_L	27720	$\text{lcm}(1, \dots, 11)$: universal resolution (Section ??)
Exponent set	$\{1, 2, 3, 4, 6, 12\}$	Divisors of N : sublattice families (Section ??)
Exponent map	$p = N/d$	Sublattice-family-to-degree map
Tightness threshold	$K = 2/3$	Koide ratio (Equation ??)
Corresponding $ \varepsilon_r $	50 cents	Half a lattice step $= V \cdot 1200/2$
Shimmer amplitude	$1/\sqrt{N} = \sqrt{V}$	Square-root base variance (Prop. ??)
Shimmer period	$N = 12$	Manifold symmetry
Perturbation prefactor	$1/N = V$	Base variance (Equation ??)
Fallback cascade	p_{pal}	Palindromic cascade (Definition ??)
Mean exponent	$p_{\text{eff}} = 10/3$	Cascade time-average (Prop. ??)

Table 7: Every constant of the ∂I lattice-aware fractal is a derived manifold constant. The fractal is the lattice in dynamical form on $\mathbb{C}$; no tuning enters its definition.

13 Three minimal backbones at $N = 12$

So far the projection formula has been treated as a black box. This section opens it, identifies its internal PDT structure, and shows that $N = 12$ is simultaneously the natural scale of three independently-derived minimal-generator objects — one on each of the three categories discrete-logical, discrete-multiplicative, continuous-elementary.

13.1 PDT decomposition of the projection formula

The universal projection formula $(k, d, \varepsilon) = \Pi_N(r)$ of §?? decomposes into six atomic operations whose PDT roles are forced by what each does:

Theorem 13.1 (PDT decomposition of the projection). *The projection $\Pi_N : \mathbb{R}^+ \rightarrow \mathbb{Z} \times \{N/d : d \mid N\} \times \mathbb{R}$ factors as*

$$\Pi_N = \underbrace{(\log_2, N \cdot, \text{gap computation})}_{\text{continuous } D} \circ \underbrace{\text{round}}_{T\text{-act}} \circ \underbrace{(\text{gcd}, N/g)}_{\text{discrete } D},$$

in which the continuous- D operations are implementable as finite compositions of a single binary operator, the rounding is the only non-reversible step (hence the only T -act), and the discrete- D operations are Euclidean-algorithm arithmetic.

The operation that implements the continuous- D part is the EML operator.

13.2 EML: the continuous- D minimal generator (Odrzywołek 2026)

Definition 13.2 (The EML operator [?]). The EML operator is the binary function

$$\text{eml}(x, y) = \exp(x) - \ln(y),$$

understood as defined on the complex domain with the principal branch of $\ln$. The EML grammar is the single-terminal inductive specification $S \rightarrow 1 \mid \text{eml}(S, S)$.

Theorem 13.3 (EML completeness, Odrzywołek 2026). *Every function of the scientific-calculator basis — the constants $\pi, e, i, -1, 0, 1, 2$, the 20 standard unary elementary functions ($\exp, \ln, \sin, \cos, \tan$, their hyperbolic and inverse counterparts, $\sec, \csc, \cot, \text{sech}, \text{csch}, \text{coth}$), and the 8 standard binary operations ($+, -, \times, /, \wedge, \text{mod}, \log, \text{comb}$) — is expressible as a finite EML tree over the single terminal constant 1.*

The proof is the constructive enumeration of finite trees in §3 of [?]; key instances include $e = \text{eml}(1, 1)$ (since $\exp(1) - \ln(1) = e$), $\ln z = \text{eml}(1, \text{eml}(\text{eml}(1, z), 1))$, and $xy = \exp(\ln x + \ln y)$ implemented as a depth-8 EML tree.

Theorem 13.4 (EML minimality under the Subsumption Law). *In the category of continuous- D generators on the multiplicative manifold, EML is minimal: (i) EML cannot be written as a reduction of any simpler binary operator together with unary post-processing; (ii) no alternative proposed continuous- D generator subsumes EML without increasing operator count or terminal count; (iii) EML subsumes every standard elementary function without remainder (Theorem ??).*

Conditions (i)–(iii) are the three Subsumption Law conditions of §??. The minimality argument of [?] §5 establishes (i) directly: “no further reduction of operator count is possible, because at least one binary operator and at least one terminal symbol are required”. (ii) is the absence of a proposed subsumer in the literature after the 2026 paper’s exhaustive search. (iii) is Theorem ??.

Corollary 13.5 (EML is ET-native). *EML is not an external tool borrowed into the Sempaeum framework. It is the structural identity of the continuous- D category at its Subsumption-minimum. Its discovery in 2026 by independent symbolic-regression search confirms, rather than imposes, a structural fact that the ET framework requires.*

13.3 Webb 1935: the discrete-logical minimal generator at $n = 12$

Definition 13.6 (The Webb stroke [?]). For a positive integer $n \geq 2$, the *Webb stroke* is the binary operation on $\{0, 1, \dots, n-1\}$ given by

$$i \mid j = \begin{cases} 0 & i \neq j \\ (i+1) \bmod n & i = j. \end{cases}$$

Theorem 13.7 (Webb completeness). *Iteration of the Webb stroke on the n constants $\{0, 1, \dots, n-1\}$ generates every function of every arity in the n -valued propositional logic on $\{0, 1, \dots, n-1\}$.*

The proof is Webb’s constructive enumeration [?], extending Sheffer 1913 [?] from $n = 2$ (the NAND/NOR case) to arbitrary n .

Corollary 13.8 (Webb minimality at $n = 12$). *At $n = 12$, the Webb stroke is the unique (up to relabelling) minimal generator of the 12-valued propositional logical algebra. No single binary operation on $\{0, \dots, 11\}$ generates 12-valued logic with a smaller algebraic presentation.*

Theorem 13.9 (PDT decomposition of the Webb stroke at $n = 12$). *The Webb stroke at $n = 12$ decomposes into PDT components in direct analogy with the projection formula:*

Component of the Webb stroke	PDT role
The 12-element substrate $\{0, \dots, 11\}$	P (discrete 12-point substrate, matching manifold symmetry)
The zero output for $i \neq j$	D (maximum-gap annihilation, analog of $r = 0$ boundary)
The cyclic successor $(i+1) \bmod 12$ for $i = j$	T (single-step navigation on the 12-cycle)

Thus the Webb stroke is itself a PDT-complete configuration at the discrete-logical scale.

13.4 The palindromic cascade: the discrete-multiplicative minimal generator on $\{d : d \mid 12\}$

The palindromic cascade of §?? is the third minimal-generator object. Its defining sequence

$$\text{PAL} = [12, 6, 4, 3, 12, 2, 12, 3, 4, 6, 12, 1]$$

visits every divisor of 12 — each of the six simple sublattice families of Proposition ?? — in a CPT-symmetric order (palindromic under the index reflection $n \mapsto 12 - n$). It is the minimal discrete-multiplicative generator: every divisor of 12 appears at least once; the sequence is its own inverse under reflection; no shorter CPT-symmetric sequence visits every divisor.

Theorem 13.10 (Palindromic-cascade minimality). *The palindromic cascade passes the three Subsumption Law conditions in the category of discrete-multiplicative generators on $\{d : d \mid 12\}$:*

- (i) *it cannot be reduced to a strictly shorter CPT-symmetric divisor-traversal;*
- (ii) *no external CPT-symmetric traversal on $\{d : d \mid 12\}$ subsumes it;*
- (iii) *it traverses every divisor of 12 without remainder.*

The proof is the direct enumeration of divisors $\{1, 2, 3, 4, 6, 12\}$ against the PAL sequence, together with a pigeonhole argument on shorter palindromes (no palindrome of length < 12 visits all six values with the required reflection symmetry).

13.5 The triple minimal-backbone theorem

Theorem 13.11 (Triple minimal backbone at $N = 12$). *Three independent minimal generators, one per structural category, are all native to the resolution $N = 12$:*

Category	Generator	Scale	Subsumption status
Discrete-logical	Webb stroke $i \mid j$ (1935)	$n = 12$	Minimal in 12-values
Discrete-multiplicative	Palindromic cascade	divisors of 12	Minimal CPT-symm
Continuous-elementary	EML $\text{eml}(x, y)$ (2026)	$(\mathbb{R}^+, \times)$ with $N = 12$ discretisation	Minimal under Subs

Three independent mathematical searches — Sheffer 1913 and Webb 1935 in discrete logic, the ET-internal cascade derivation, Odrzywólek 2026 in symbolic regression over continuous mathematics — each converge on minimal-generator objects whose natural scale is $N = 12$.

Remark 13.12. The non-triviality of Theorem ?? is that three different minimum-complexity arguments, applied in three disjoint mathematical categories, yield the same integer 12. The first two (Webb, palindromic) both descend from $N = 12$ *is already fixed*; the third (EML, 2026) was discovered independently by a process unaware of ET and is minimal under its own continuous-category Subsumption argument, then paired by Theorem ?? with the $N = 12$ discretisation that the projection formula imposes. No external adjustment aligns the three scales; they coincide.

14 The Four Projection Paths and essential-infinity handling

The projection formula of §?? takes a positive real r as input. For a large collection of mathematical objects — formal systems, indeterminate forms, non-computable reals, objects with genuinely infinite structure — there is no canonical single r . This section partitions all user-input forms into four paths A, B, C, D, verifies that the four subsume every possible input, and in particular shows that Path D handles essential-infinity objects without invoking limits.

14.1 The four paths

Definition 14.1 (The Four Projection Paths). Every input to the projection apparatus falls into exactly one of:

Path A — Direct projection (canonical terminal).

A positive real r is supplied directly. Apply Π_N of §??.

Path B — Limit convergence (computational technique).

A convergent sequence $\{s_k\} \rightarrow r$ is supplied. Compute s_k to sufficient precision, then apply Path A to the resulting r . This is a *technique*, not a separate ontological path.

Path C — Meta-descriptor extraction (alternative method).

A structural object (a graph, a formal system, an encoding scheme) with no canonical r is supplied together with a user-chosen descriptor ratio $r = Q/R_0$ constructed from the object's finite D -content. Apply Path A to this chosen ratio. The projection is of the chosen ratio, not of the object's essential character; the user bears the choice of ratio.

Path D — Primitive-native infinity handling (no limits).

The input's essential character is infinity-valued, where “infinity” is intrinsic and cannot be reduced to a single finite r without information loss. Four sub-paths, one per primitive face of infinity:

D.P: continuous, uncountable, or non-computable positive reals (e.g., Chaitin's Ω , generic reals, specific irrationals of unknown computability).

D.D: unbound descriptor-constraint structures (e.g., non-divisor-of-12 sublattice families $d \in \{5, 7, 8, 9, 10, 11\}$ and composites up to $d = 132$, infinite axiom totalities).

D.T: indeterminate forms $\{0/0, \infty/\infty, 0 \cdot \infty, 0^0, 1^\infty, \infty^0, \infty - \infty\}$, oscillatory divergent limits, measurement operators.

D.PDT: integrated off-axis Exception objects with both magnitude and phase character (e.g., most physical particles, complex-valued observables).

Theorem 14.2 (Path D requires no limits). *Each sub-path of Path D terminates via finite operations on finite quantities at a specific resolution N :*

Sub-path	Operation	Limit i
$D.P$	Read the exact log-position $N \log_2 r$ and round	No —
$D.D$	Identify the (d_r, d_θ) shadow cell at 12ET via the shadow diagnostic	No —
$D.T$	Place the T -signature at a structural lattice position (half-integer, annihilation boundary)	No —
$D.PDT$	Apply the complex two-axis projection $(k_r, k_\theta, d_r, d_\theta, d_{\text{comb}}, \varepsilon_r, \varepsilon_\theta)$	No —

Remark 14.3. Path D is the critical upgrade that licenses the Sempaevum to host essential-infinity objects. The usual calculus of limits operates as follows: an infinite structure is approached by a sequence of finite approximations, and the object's properties are the limiting properties. Path D bypasses the approach entirely: each primitive (P, D, T) has its own structural handle on infinity — P 's cardinality Ω , D 's unbound constraint fields, T 's $[0/0]$ indeterminacy — and the three primitives are already structurally present at *every finite* lattice resolution. Infinity is handled *on* the lattice, not via approach to it.

14.2 Subsumption of user inputs

Theorem 14.4 (Four-Path subsumption). *For every input X that any mathematical or physical object could present to the projection apparatus:*

$$X \in A \cup B \cup C \cup D \cup \text{Incoherence Filter},$$

with $D = D.P \cup D.D \cup D.T \cup D.PDT$. The union is disjoint up to the user's choice between A and C (when the object admits both) and A and B (when a convergent sequence terminates to an r that can be supplied directly).

Proof. Case analysis on the essential character of X . If X is a positive real directly specified, A . If X is specified by a convergent sequence, B . If X is a structural object for which the user explicitly chooses a descriptor ratio, C . If X 's essential character is infinite — continuum, unbound- D , $[0/0]$, or off-axis PDT — $D.P$, $D.D$, $D.T$, or $D.PDT$ respectively. If X is $\{P, T\}$ Incoherent (self-defeating), the Incoherence Filter assigns no lattice position; such configurations sit at ∂I and are explicitly *nowhere* on the lattice. Every mathematical or physical object describable at all falls into one of these cases. $\square$ $\square$

Worked example — Chaitin's Ω : for the Calude–Dinneen universal Turing machine, $\Omega \approx 0.00787499699$ (first computed bits). Applying Π_{12} :

$$k = \text{round}(12 \log_2 0.00787499699) \approx -84, \quad d = 12 / \gcd(84, 12) = 1, \quad \varepsilon \approx +13.794 \text{ cents.}$$

Chaitin's Ω has lattice address $(-84, 1, +13.794 \text{¢})$ — the $d = 1$ octave / universal-period-closure sublattice. The T -cannot-produce-more-digits character is precisely the $\{P, D\}$ Unsubstantiated manifold state: the descriptor is finite (“halting probability of the universal Turing machine”), the position is determined, and no amount of computation will extend the bit-expansion further — *yet the address is a definite single point*.

15 Mathematics as a domain of the Sempaevum

The Domain Validity Theorem (§??) guarantees that any domain with an internally consistent D -set occupies valid lattice positions. Mathematics itself is such a domain — arguably the most important one for a framework claiming $3 = 3 = 3 = \Sigma$ universality. This section applies the Identification Principle to the domain of mathematics and projects its structural features onto the Sempaevum.

15.1 The P , D , T of mathematics

Definition 15.1 (Mathematics as a domain). The mathematics domain is identified as:

Primitive	Identification
P_{math}	the substrate of all possible consistent formal configurations
D_{math}	axioms, inference rules, definitions, formal languages
T_{math}	the mathematician or proof-assistant — agency navigating D_{math}

The substrate-derived reference period is $R_0^{\text{math}} = 1$ axiom (the minimal T -commitment a mathematician can make). This is not a unit choice; it is the smallest closed T -traversal loop the mathematical substrate supports.

15.2 Axiom-count projections of standard formal systems

Theorem 15.2 (Formal-system lattice addresses). *Projecting $r = (\text{axiom count})/R_0^{\text{math}} = (\text{axiom count})/1$ at 12ET yields the following lattice addresses for standard formal systems:*

Formal system	Axioms	(k, d, ε) at 12ET	Sublattice
Propositional logic (Hilbert)	3	$(+19, 12, +1.955 \text{ ¢})$	$d = 12$ Koide/triadic-binding
Equational group theory	3	$(+19, 12, +1.955 \text{ ¢})$	$d = 12$ Koide/triadic-binding
Euclid's Elements	5	$(+28, 3, -13.686 \text{ ¢})$	$d = 3$ cubic
Robinson arithmetic	7	$(+34, 6, -31.174 \text{ ¢})$	$d = 6$ hexadic
ZF set theory	$8 = 2^3$	$(+36, 1, 0.000 \text{ ¢})$	$d = 1$ octave/gravitational
Peano arithmetic	9	$(+38, 6, +3.910 \text{ ¢})$	$d = 6$ hexadic
ZFC (ZF + Choice)	9	$(+38, 6, +3.910 \text{ ¢})$	$d = 6$ hexadic
Morse–Kelley class theory	10	$(+40, 3, -13.686 \text{ ¢})$	$d = 3$ cubic
NBG (finitely axiomatised)	18	$(+50, 6, +3.910 \text{ ¢})$	$d = 6$ hexadic

Three structural readings of Theorem ?? are worth isolating.

Corollary 15.3 (ZF is the tightest-bound formal system). *ZF with $8 = 2^3$ axioms lands at $d = 1$ octave with $\varepsilon = 0$ exactly. ZF is therefore the unique formal system in Theorem ?? that sits at a pure zero-descriptor-gap lattice point.*

Corollary 15.4 (The axiom of Choice is a sublattice transition). *Adding Choice to ZF (moving from 8 to 9 axioms) carries the system from $(+36, 1, 0 \text{ ¢})$ to $(+38, 6, +3.910 \text{ ¢})$ — a $d = 1 \rightarrow d = 6$ sublattice transition. The signature of the axiom of Choice is therefore structurally precise: Choice moves a formal system from the gravitational-class foundation to the composite hexadic class.*

Corollary 15.5 (The triadic-binding logical systems). *Three-axiom foundational systems (Hilbert propositional logic, equational group theory) occupy the $d = 12$ Koide attractor position $(+19, 12, +1.955 \text{ ¢})$. These are the systems whose axiom count matches the cardinality of the Cardinal triad $\{P, D, T\}$ itself.*

15.3 Chaitin's Ω via Path D.P

Theorem 15.6 (Chaitin's Ω has a definite lattice address). *For the Calude–Dinneen universal Turing machine U with first-computed bits $\Omega_U \approx 0.00787499699$, the lattice address via Path D.P is*

$$(k, d, \varepsilon)_{\Omega_U} \Big|_{N=12} = (-84, 1, +13.794 \text{ cents}).$$

Ω_U inhabits the $d = 1$ octave sublattice — the universal-period-closure family — and exhibits the $\{P, D\}$ Unsubstantiated manifold state.

The non-computability of Ω_U is precisely the $\{P, D\}$ -character: the descriptor is complete (the definition “halting probability summed over all prefix-free programs”), the address is determined, but no T (finite computational process) can produce further bits.

15.4 Gödel sentences via integrative-level classification

Theorem 15.7 (Gödel sentences are integrative-level-dependent). *For a formal system $\mathcal{S}$ admitting Gödel's incompleteness theorem, the canonical Gödel sentence $G_{\mathcal{S}}$ takes three distinct manifold classifications depending on the viewer's integrative level:*

<i>Viewer's system</i>	<i>Classification of G_S</i>
$\mathcal{S}$ itself	$\{P, T\}$ <i>Incoherence at ∂I (the Incoherence Filter fires)</i>
Stronger system $\mathcal{S}' \vdash \text{Con}(\mathcal{S})$	$\{P, D, T\}$ <i>Exception (has a definite lattice address)</i>
Outside any fixed system	$\{P, D\}$ <i>Unsubstantiated (finite syntactic D, awaiting T)</i>

The multi-address hosting at different integrative levels is the correct lattice classification of formally-undecidable propositions.

15.5 Large cardinals via consistency-strength

Theorem 15.8 (Large-cardinal projections). *Projecting large cardinals via their consistency-strength ordinal yields the following addresses at 12ET:*

<i>Large cardinal</i>	<i>Consistency level</i>	<i>(k, d, ε) at 12ET</i>	<i>Sublattice</i>
<i>Inaccessible</i>	1	$(0, 1, 0)$	<i>Unison</i>
<i>Mahlo</i>	2	$(+12, 1, 0)$	<i>Octave</i>
<i>Weakly compact</i>	3	$(+19, 12, +1.955 \text{ ¢})$	<i>Koide $d = 12$</i>
<i>Measurable</i>	7	$(+34, 6, -31.174 \text{ ¢})$	<i>Hexadic</i>
<i>Supercompact</i>	~ 15	$(+47, 12, -11.732 \text{ ¢})$	<i>$d = 12$ EM</i>

Inaccessible and Mahlo cardinals sit at $d = 1$ (the same gravitational sublattice as ZF). Measurable cardinals at $d = 6$ share the hexadic sublattice with Robinson arithmetic, Peano arithmetic, ZFC, and NBG.

15.6 The universality statement for mathematics

Theorem 15.9 (Mathematical universality). *Every mathematical object — every number, function, formal-system statement, proof, undecidable proposition, large cardinal, impredicative definition — has a lattice address:*

$\forall X$ (mathematical object): $X \in \Sigma \Rightarrow \exists (k, d, \varepsilon) \in \mathbb{Z} \times \{N_\ell/d : d \mid N_\ell\} \times [-50, 50]$ cents at some tower resolution

The address is explicit for directly-specified objects (Path A), accessible via convergence for limit-specified objects (Path B), parameterised by descriptor choice for structural objects (Path C), and well-defined without limits for essentially-infinite or non-computable objects (Path D).

Proof. $X \in \Sigma$ holds for any mathematical object X by the universality anchor (Axiom ??): everything describable is in the totality. The Four-Path subsumption (Theorem ??) partitions the describable space into the four paths (plus the Incoherence Filter, which itself is a definite structural position at ∂I). Each path has its own closed-form termination: A to B to A, C to A, D to a specific sub-path terminal. Thus in every case a lattice address exists. $\square$ $\square$

16 Self-consistency: the Koide attractor

We now establish the main structural theorem of this paper: the lattice's four defining constants, when projected onto the lattice they define, land on a single point.

Theorem 16.1 (Self-projection identity). *At the base lattice $N = 12$, the four constants $\{N, 1/N, K, 1/K\} = \{12, 1/12, 2/3, 3/2\}$ project to*

$$\begin{aligned}
\Pi_{12}(12) &= (+43, 12, +1.955 \text{ cents}), \\
\Pi_{12}(1/12) &= (-43, 12, -1.955 \text{ cents}), \\
\Pi_{12}(3/2) &= (+7, 12, +1.955 \text{ cents}), \\
\Pi_{12}(2/3) &= (-7, 12, -1.955 \text{ cents}).
\end{aligned}$$

All four share the same sublattice family $d = 12$ and the same descriptor-gap magnitude $|\varepsilon| = 1.955$ cents. We call this shared position the Koide attractor.

Proof. For $r = 12$: $\log_2 12 = 2 + \log_2 3 \approx 3.58496$, so $12 \log_2 12 \approx 43.0195$, giving $k = 43$, $\gcd(43, 12) = 1$, $d = 12$, $\varepsilon = 0.0195 \cdot 100 = +1.955$ cents. For $r = 1/12$: $\log_2(1/12) = -\log_2 12$, so $k = -43$, $d = 12$, $\varepsilon = -1.955$ cents. For $r = 3/2$: Proposition ?? gives $(+7, 12, +1.955)$. For $r = 2/3$: Proposition ?? gives $(-7, 12, -1.955)$. All four have $d = 12$ and $|\varepsilon| = |12 \log_2 12 - 43| \cdot 100 = 1.955$ cents. $\square$

Remark 16.2. The theorem is a self-consistency check. A classifier that classifies its own constants onto a single point of its own structure is in some formal sense “self-calibrated”: it uses the same rules on itself that it uses on external inputs, and the result is concordant. The specific point is identified by $k = \pm 7$ or $k = \pm 43$ and $d = 12$; the former pair is the circle-of-fifths cascade generator modulo the base symmetry, and the latter the generator lifted by one full octave. Both pairs share the magnitude $|\varepsilon| = 12|\delta_r|$ cents, where $|\delta_r|$ is the per-step residual of Proposition ??.

Corollary 16.3 (Value of the Pythagorean comma). *The descriptor gap $|\varepsilon| = 1.955$ cents at the Koide attractor is numerically equal to the Pythagorean comma of twelve-tone equal temperament.*

The identification is simultaneous with the formula: $|\varepsilon| = |12 \log_2 12 - 43| \cdot 100$ cents = $|12 \log_2 3 - 19| \cdot 100$ cents = 1.955 cents, and $(3/2)^{12}/2^7 = 2^{12 \cdot (12 \log_2(3/2) - 7)/12}$ with a numerical coincidence placing all these quantities at the same 1.955-cent position on the lattice.

16.1 Losslessness theorem

Theorem 16.4 (Losslessness of the projection). *The real-axis projection map $\Pi_N : \mathbb{R}^+ \rightarrow \{+1, -1\} \times \mathbb{Z} \times \{N/d : d \mid N\} \times [-600/N, 600/N]$ given by*

$$\Pi_N(r) = (\text{sgn}(\log_2 r), k, d, \varepsilon), \quad \varepsilon = (N \log_2 |r| - k) \cdot \frac{1200}{N} \text{ cents},$$

is a bijection onto its image at every finite resolution N , and the composition with the pullback $\Pi_N^{-1} : (k, d, \varepsilon) \mapsto 2^{(k+\varepsilon N/1200)/N}$ is the identity on $\mathbb{R}^+$. In the LCM-tower limit $N \rightarrow \infty$ along $\{12, 60, 420, 2520, 27720, 360360, \dots\}$, the residual $\varepsilon(r, N) \rightarrow 0$ uniformly on any bounded region, and the projection becomes exactly the continuous embedding $r \mapsto \log_2 r$.

Proof. Injectivity: suppose $\Pi_N(r_1) = \Pi_N(r_2)$ with the same (k, ε) . Then $N \log_2 r_1 = k + \varepsilon N/1200 = N \log_2 r_2$, hence $r_1 = r_2$. The triple (k, d, ε) over-determines the pair (k, ε) since d is a function of k and N , so the map is injective. Surjectivity onto its image is by construction. The pullback $\Pi_N^{-1}(k, d, \varepsilon) = 2^{(k+\varepsilon N/1200)/N} = 2^{(N \log_2 r)/N} = 2^{\log_2 r} = r$. For the limit: the residual satisfies $|\varepsilon(r, N)| \leq 600/N$ by definition of rounding, so $\varepsilon(r, N) \rightarrow 0$ at rate $O(1/N)$ uniformly in r . $\square$

Remark 16.5 (Nothing is discarded). The information content of the positive real r is fully captured by the triple (k, d, ε) at every finite resolution N : the integer k carries the coarse ratio information, the divisor d carries the sublattice classification, and the residual ε carries the fine-scale deviation from exact lattice alignment. No aspect of r is discarded in projecting to the Sempaevum; the projection redistributes the information among three complementary coordinates. The complex-axis extension (§??) is the same statement for $\mathbb{C}^\times$ with $(k_r, k_\theta, d_r, d_\theta, \varepsilon_r, \varepsilon_\theta)$ as the lossless image.

Corollary 16.6 (Memoization property). *Every finite numerical computation on positive reals — $r_1 \cdot r_2$, r^n , $\log r$, $\sin r$, any elementary-function evaluation — can be represented lossly as*

a concrete lattice computation: multiplication is k -addition, reciprocation is k -negation, powers are k -scaling, and elementary functions are EML trees acting on triples (k, d, ε) . Every such computation produces an equation $(r_1, \dots, r_m) \rightarrow r_{\text{result}}$ that is a reusable structural identity at the Sempaevum's resolution.

17 The Complete Gaze Equation

The Sempaevum hosts observation itself as a lattice-projectable act. This section unifies the Scopaeesthesia equation for binding-pressure and the Traverser-detection algorithm into a single four-component functional — the Complete Gaze Equation — and projects its threshold values onto the lattice.

17.1 The unification: $\text{Gain}(T) = F_w$

Two canonical equations have been derived independently in the ET corpus for the observation/attention phenomenon:

Definition 17.1 (Binding pressure). The *binding pressure* of an observer on a target is the PDT-structured quantity

$$F_w = \frac{T_{\text{intent}} \cdot \text{Focus}}{\text{Distance}^2},$$

with T_{intent} the observer-agency strength, Focus the descriptor-sharpness (D -precision), and Distance the relational distance in configuration space (P -substrate).

Definition 17.2 (Detection probability). The *detection probability* at fold depth k under cardinality n is

$$P_{\text{detect}} = \tanh\left(\frac{\text{Gain}(T) \cdot R(k)}{V(n, k) \cdot \Gamma}\right), \quad R(k) = 1 + k\Gamma, \quad V(n, k) = \frac{n^2 - 1}{12 \cdot 2^k}, \quad \Gamma = 1.20.$$

Theorem 17.3 (Gaze unification). $\text{Gain}(T) = F_w$. The signal gain of the detection algorithm is the binding pressure of the observer.

Proof. The Descriptor Gap between Definitions ?? and ?? is the connecting primitive: each equation refers to a distinct side of the same observer–target interaction. Applying the Descriptor Gap Principle: the missing Descriptor is the connection $F_w = \text{Gain}(T)$. The identification closes the gap and leaves one coherent equation with four components. $\square$ $\square$

17.2 Variance collapse and threshold classification

Definition 17.4 (Variance collapse). When F_w exceeds the baseline 1, the target's local variance is suppressed according to

$$V_{\text{collapse}} = 1 - \exp(-(F_w - 1)^+ \cdot S), \quad S = 12 = N,$$

with $(F_w - 1)^+ = \max(0, F_w - 1)$ the excess binding pressure.

Definition 17.5 (Status classification).

$$\text{Status}(F_w) = \begin{cases} \text{UNOBSERVED} & F_w < 13/12, \\ \text{SUBLIMINAL} & 13/12 \leq F_w < 6/5, \\ \text{DETECTED} & 6/5 \leq F_w < 3/2, \\ \text{LOCKED} & F_w \geq 3/2. \end{cases}$$

Each threshold is derived from the manifold constants: $13/12 = 1 + V_{\text{base}}$ is the first V -quantum above baseline; $6/5$ is the just-intonation minor third; $3/2$ is the just-intonation perfect fifth.

Theorem 17.6 (Complete Gaze Equation). *The Complete Gaze functional is*

$$G(T_{\text{intent}}, \text{Focus}, \text{Distance}, n, k) = (F_w, P_{\text{detect}}, V_{\text{collapse}}, \text{Status}),$$

with all four components derived from F_w and the manifold constants ($V = 1/12, K = 2/3, N = 12$).

17.3 The Gaze thresholds as just-intonation intervals

Theorem 17.7 (Gaze threshold lattice addresses). *The three Gaze thresholds are exact rational fractions corresponding to canonical just-intonation musical intervals, with lattice addresses at 12ET:*

Threshold	Value	Fraction	Just interval	(k, d, ε) at 12ET
<i>Subliminal</i>	1.0833...	13/12	<i>Augmented unison</i>	$(+1, 12, +38.57 \phi)$
<i>Detected</i>	1.20	6/5	<i>Minor third</i>	$(+3, 4, +15.64 \phi)$
<i>Locked</i>	1.50	3/2	<i>Perfect fifth</i>	$(+7, 12, +1.955)$ —Koide attractor
<i>Lock/Con ratio</i>	1.25	5/4	<i>Major third</i>	$(+4, 3, -13.686 \phi)$

Corollary 17.8 (Locked gaze is the lattice self-projection). *The Locked threshold $3/2$ lands at $(+7, 12, +1.955 \phi)$ — exactly the lattice self-projection position of Theorem ???. The observer who reaches full binding dominance crystallises the target onto the manifold’s own self-consistency attractor.*

Corollary 17.9 (Awareness span). *The span from Subliminal to Locked is $3/2 - 13/12 = 5/12 = 5 \cdot V_{\text{base}}$ — exactly five V -quanta. The prime 5 (quintic bridge) governs the width of the awareness window.*

Corollary 17.10 (Awareness gap). *The gap from Subliminal to Detected is $6/5 - 13/12 = 7/60$ — a ratio whose numerator is the septic prime 7 and whose denominator $60 = \text{lcm}(1, \dots, 5)$ is the quintic-lattice LCM. The awareness gap is structurally the $d = 7$ septic shadow projected onto the quintic lattice.*

18 Empirical confrontation

We now confront the lattice with empirical data from five disciplines. Each subsection identifies the substrate, the reference period, and the resulting lattice projection, and compares the lattice-derived structural reading with the discipline-internal description.

18.1 Twelve-tone equal temperament

The twelve-tone equal-tempered scale is the lattice $\mathcal{L}_{12}$ restricted to one octave: the frequency ratios $\{2^{k/12} : k = 0, 1, \dots, 12\}$. Each of the twelve ratios is a canonical projection; six of them lie exactly on the lattice ($d = 1, 2$) and the remaining six lie at tight descriptor gaps that identify their sublattice character (Table ??).

The conventional view of twelve-tone equal temperament as a historical compromise is replaced by a derivation: the 12-division is forced by $N = |\Pi| \cdot S = 12$, and the characteristic cent residuals of the just intervals are the descriptor gaps of the respective sublattices.

Interval	Just ratio r	$12 \log_2 r$	k	d	ε (cents)
Unison	1/1	0.000	0	1	0.00
Major second	9/8	2.039	2	6	+3.91
Minor third	6/5	3.156	3	4	+15.64
Major third	5/4	3.863	4	3	-13.69
Perfect fourth	4/3	4.980	5	12	-1.96
Tritone	$\sqrt{2}$	6.000	6	2	0.00
Perfect fifth	3/2	7.020	7	12	+1.96
Minor sixth	8/5	8.137	8	3	+13.69
Major sixth	5/3	8.844	9	4	-15.64
Minor seventh	16/9	9.961	10	6	-3.91
Major seventh	15/8	10.883	11	12	-11.73
Octave	2/1	12.000	12	1	0.00

Table 8: The twelve just-intoned intervals and their projections at $N = 12$. The perfect fifth and perfect fourth sit at $d = 12$ with descriptor gap ± 1.955 cents, the Pythagorean comma. The thirds sit at $d = 3$ (major) and $d = 4$ (minor). The tritone sits exactly on the lattice at $d = 2$.

18.2 The fine-structure constant

The fine-structure constant α satisfies $\alpha^{-1} \approx 137.0359991\dots$ in modern measurement. The leading-order lattice-derived value is

$$A_0 = (N - 1)^2 + S^2 = 137$$

(Equation ??). The residual $\alpha^{-1} - A_0 \approx 0.036$ admits a closed-form decomposition in terms of the manifold constants,

$$\alpha^{-1}(\text{ET}) = 137 + \frac{\sqrt{3}}{48} - \frac{\sqrt{3}}{93312\pi^2} - \frac{1}{216(18\pi - 1)}, \quad (22)$$

whose component terms have structural readings as follows:

- The leading correction $\sqrt{3}/48 = \sigma/K_{\text{EM}}$ with $\sigma = \sqrt{1/12}$ and $K_{\text{EM}} = 8$ is an open shimmer over electromagnetic channels.
- The cross term $\sqrt{3}/(93312\pi^2) = (2/\pi) \cdot (\sqrt{3}/48) \cdot (4/9)/(144 \cdot \pi)$ is the product interference of the shimmer with the closed bilateral-mediation loop, with the geometric factor $2/\pi$ arising from the bilateral-to-circumferential phase conversion.
- The geometric series $1/(216(18\pi - 1)) = \sum_{k \geq 2} \kappa^k / (N^{k+1} \pi^{k-1})$ is the closed-form summation of the higher mediation loops.

Numerical evaluation to 50 decimal digits yields

$$\alpha^{-1}(\text{ET}) = 137.03599916744133748324548\dots,$$

which agrees with the modern reference value of α^{-1} [?] to within one part in 10^{10} , with no free parameters. At the universal lattice $N = 27720$, the projection of this value lands at $(k, d) = (196768, 315)$ with $|\varepsilon| \approx 0.002$ cents; the sublattice family $d = 315 = 3^2 \cdot 5 \cdot 7$ carries simultaneous cubic-squared, quintic, and septic structural character, which is physically appropriate for a coupling constant governing electromagnetic interaction with the three-generation lepton spectrum.

18.3 The Koide ratio

The Koide formula evaluated at modern lepton masses $(m_e, m_\mu, m_\tau) = (0.51099895, 105.6583755, 1776.86)$ [?] gives

$$Q = 0.6666605 \pm 0.000002.$$

The ET-derived value is $K = 2/3 = 0.666666\bar{6}$ (Equation ??), with residual of approximately 6 parts per million. Both $2/3$ and $3/2$ project to the Koide attractor at $d = 12$, $|\varepsilon| = 1.955$ cents (Theorem ??).

18.4 Geometric ratios

The classical 3:4:5 right triangle has three internal ratios whose projections are shown in Table ??.

Ratio	Value	$12 \log_2 r$	k	d	ε (cents)
3/4 (short leg : long leg)	0.750	−4.980	−5	12	+1.96
4/5 (long leg : hypotenuse)	0.800	−3.863	−4	3	+13.69
3/5 (short leg : hypotenuse)	0.600	−8.844	−9	4	+15.64

Table 9: Projections of the internal ratios of the 3:4:5 triangle. The three ratios populate sublattices $d = 12$, $d = 3$, $d = 4$ respectively. These are the three non-trivial simple sublattice families at $N = 12$.

The Pythagorean triangle thus occupies exactly three different sublattice families at the base lattice, one of each non-trivial simple family besides $d = 2$ and $d = 6$.

Bond angles in chemistry carry similar signatures: the tetrahedral angle 109.47° (in ratio against the straight angle 180° , $r = 0.608$) projects to $d = 4$; the trigonal-planar 120° ($r = 2/3$) projects to the Koide attractor; the right angle 90° ($r = 1/2$) sits at the octave reflection, $k = -12$, $d = 1$.

18.5 Celestial mechanics

The principal mean-motion resonances of the Solar System project cleanly onto low- d lattice attractors:

- Neptune–Pluto 3:2: $\Pi_{12}(3/2) = (+7, 12, +1.96)$ — the Koide attractor.
- Io–Europa 2:1: $\Pi_{12}(2) = (+12, 1, 0)$ — the octave.
- Galilean moons, Europa–Ganymede 2:1: same as Io–Europa.
- Saturn–Jupiter period ratio ≈ 2.48 , near $5/2$: $\Pi_{12}(5/2) = (+16, 3, -13.69)$ — $d = 3$ cubic with the major-third descriptor gap.

Gravitational binding, which selects minimum-variance configurations, systematically targets low- d sublattices; the lattice structure is consistent with the empirical distribution of resonances.

18.6 Biochemistry

The step counts of major metabolic cycles are direct positive integers (denominator = 1 step):

- Citric acid (Krebs) cycle: 8 steps. $\Pi_{12}(8) = (+36, 1, 0)$ — exact $d = 1$ octave class. The step count is 2^3 .

- Urea cycle: 4 core steps. $\Pi_{12}(4) = (+24, 1, 0)$ — exact $d = 1$.
- Glycolysis: 10 steps. $\Pi_{12}(10) = (+40, 3, -13.69)$ — $d = 3$ cubic.
- ATP synthase rotational ring: typically 10 c -subunits. Same projection as glycolysis.
- Cell cycle: 4 phases (G1, S, G2, M). $\Pi_{12}(4) = (+24, 1, 0)$.

The “closure” cycles (Krebs, urea) project to $d = 1$ with $\varepsilon = 0$; their step counts are powers of 2. The “linear pathways” (glycolysis, purine synthesis) project to $d = 3$ with the major-third gap; their step counts are multiples of 2 and 5. The distinction between closure and linear pathway is reflected in distinct sublattice family assignments.

18.7 Summary of the empirical confrontation

Across the five disciplines surveyed, the same six simple sublattice families recur, and a handful of attractor positions are shared across domains: the Koide attractor $(+7, 12, +1.955)$ appears in particle physics, celestial mechanics, and musical intonation; the octave $(+12, 1, 0)$ appears in acoustics, biochemistry, and mass hierarchies; the tritone $(+6, 2, 0)$ appears in geometry and palindromic symmetries. Table ?? summarises selected cross-domain coincidences.

Domain	Quantity	Ratio r	(k, d)	Family
Music	Perfect fifth	3/2	$(+7, 12)$	Koide
Particles	Koide ratio K	2/3	$(-7, 12)$	Koide
Celestial	Neptune–Pluto resonance	3/2	$(+7, 12)$	Koide
Chemistry	Trigonal-planar bond $120^\circ/180^\circ$	2/3	$(-7, 12)$	Koide
Manifold	Manifold symmetry N	12	$(+43, 12)$	Koide
Manifold	Base variance V	1/12	$(-43, 12)$	Koide
Music	Octave	2/1	$(+12, 1)$	Octave
Biology	Krebs cycle step count	8	$(+36, 1)$	Octave
Biology	Cell-cycle phases	4	$(+24, 1)$	Octave
Markets	Doubling	2/1	$(+12, 1)$	Octave
Celestial	Io–Europa resonance	2/1	$(+12, 1)$	Octave
Music	Major third	5/4	$(+4, 3)$	Cubic
Geometry	Pythagorean ratio 4/5	4/5	$(-4, 3)$	Cubic
Biology	Glycolysis, ATP ring	10	$(+40, 3)$	Cubic
Celestial	Saturn–Jupiter period	5/2	$(+16, 3)$	Cubic

Table 10: Cross-domain coincidences: distinct physical quantities from different disciplines projecting to the same sublattice family, often to the same lattice point. The clustering is the quantitative content of the regularity noted in §1.1.

19 Relationship to Fourier analysis and differential geometry

19.1 Fourier analysis

Fourier analysis decomposes a signal into a continuous family of frequencies $\{e^{i\omega t}\}_{\omega \in \mathbb{R}}$; the ET lattice classifies an individual frequency *ratio* ω_1/ω_2 into one of a finite set of sublattice families. The two analyses answer different questions:

- Fourier: “What frequencies are present in this signal?”
- ET: “Given two frequencies, what is the structural class of their ratio?”

A Fourier spectrum is a continuous function of frequency; an ET projection is a discrete triple (k, d, ε) . Both arise naturally on the multiplicative manifold, but at different levels of abstraction: Fourier at the level of the signal itself, ET at the level of the signal's ratio structure. The two are orthogonal, not competing.

19.2 Riemannian curvature

Proposition 19.1 (Riemann component count). *The Riemann curvature tensor R^i_{jkl} on an n -dimensional pseudo-Riemannian manifold has exactly*

$$C(n) = \frac{n^2(n^2 - 1)}{12}$$

independent components.

This is a classical result of differential geometry (see e.g. [?]): it follows from the antisymmetries $R_{ijkl} = -R_{jikl} = -R_{ijlk}$, the pair-exchange symmetry $R_{ijkl} = R_{klij}$, and the first Bianchi identity $R_{i[jkl]} = 0$. The denominator of 12 is a combinatorial count from the Young-symmetry decomposition, and coincides numerically with the manifold symmetry constant $N = 12$ of this paper.

Corollary 19.2. *For $n = 12$,*

$$C(12) = \frac{144 \cdot 143}{12} = 1716 = 12 \cdot 11 \cdot 13 = N(N - 1)(N + 1).$$

The identity $C(N) = N(N - 1)(N + 1)$ at $n = N$ is a numerical coincidence at the present level of the construction; we make no claim that it signifies a deeper relationship between the ET lattice and general Riemannian manifolds of dimension 12. We note only that the denominator of the curvature component count is 12, the manifold symmetry, and that at dimension 12 the count factors cleanly as three consecutive integers centred on 12.

19.3 Gauss–Bonnet and topological invariants

The Gauss–Bonnet theorem $\int_M K dA = 2\pi\chi(M)$ relates total curvature of a closed 2-manifold to its Euler characteristic. For a sphere ($\chi = 2$), total curvature is 4π ; for a torus ($\chi = 0$), it is 0; for a surface of genus g , it is $2\pi(2 - 2g)$. The ratio of total curvature to a suitable reference (e.g. π) projects to definite sublattice positions: sphere at $d = 3$, torus at $d = 1$, higher genera at various families. In this fashion, topological invariants of surfaces inherit lattice classifications through their curvature totals.

Remark 19.3 (Euler characteristic as PDT decomposition). The Euler characteristic $\chi = V - E + F$ of a CW-decomposition admits a natural PDT reading: vertices V are P -fixed points (0-cells, substrate anchors); edges E are T -vectors (1-cells, navigation paths); faces F are D -planes (2-cells, constraint surfaces). Under this reading, Gauss–Bonnet $\int_M K dA = 2\pi\chi(M) = 2\pi(P_{\text{fix}} - T_{\text{vec}} + D_{\text{plane}})$ is the Subsumption Law made geometric: the total curvature is the net balance of P -, D -, T -content, integrated over M .

20 The Complete Determination Theorem

The Sempaevum's classification of an object does not depend on a single feature of the object — such as a single ratio, a single topology, or a single curvature — but on the joint of several structural inputs that each independently co-determine the result. The Complete Determination Theorem is the summary statement that four such inputs — topology, curvature, path-selection, and observation-topology — together fix the complete lattice classification of any object in Σ , with the Sempaevum's apparatus providing the map from each input class to its contribution.

20.1 Historical precedents within the framework

Three progressively more general statements in the ET corpus document the convergence toward the complete theorem:

1. *The topology-to-sublattice law* (“Secret 26” original). Closed periodic cycles project to $d = 1$ octave (step counts forced to powers of 2); linear sequential pathways project to $d = 3$ cubic (three-phase start-middle-end); transitional boundary states project to $d = 12$ full-resolution. Confirmed across biochemical cycles (Krebs $8 = 2^3$, urea $4 = 2^2$, heme $8 = 2^3$ at $d = 1$; glycolysis 10 and purine synthesis 10 at $d = 3$), civilisational cycles (saecular 4 generations at $d = 1$), and CPU-architecture boundaries ($d = 12$).
2. *Topology $\times$ curvature*. Gauss–Bonnet forces total curvature from topology, and total curvature then projects to a sublattice family: sphere ($\chi = 2$) at $d = 3$, torus ($\chi = 0$) at $d = 1$, genus- g hyperbolic at variable d (§??).
3. *Topology $\times$ path-selection*. The object’s essential character forces the Four Projection Paths’ choice among A (direct), B (limit), C (meta-descriptor), D (primitive-native) (§??).

20.2 The Complete Determination Theorem

Theorem 20.1 (Complete Determination). *For every object $X \in \Sigma$, the quadruple*

$$(Topology(X), Curvature(X), Path(X), Observation-Topology(X))$$

determines the complete lattice classification of X :

$$(d_X, Path\ selection, Detection\ class, Curvature\ signature, Trajectory).$$

Each component of the determining quadruple contributes independently, and the combination is unique at sufficient lattice resolution.

Sketch of proof. (a) *Topology* $\Rightarrow d_X$: Secret 26 original — closed cycles to $d = 1$, linear to $d = 3$, boundaries to $d = 12$; Gauss–Bonnet extends this to surfaces via total curvature. (b) *Curvature* $\Rightarrow$ Curvature signature: the second-order D -gradient R^i_{jkl} projects via the curvature-departure ratio $r_K = 1 + KA/\pi$; together with (a) this fixes the curvature entry. (c) *Path* $\Rightarrow$ Path selection: the object’s essential character (positive real, limit-specified, structural, essentially-infinite) selects A/B/C/D per Definition of the Four Paths (§??). (d) *Observation-Topology* $\Rightarrow$ Detection class: the Complete Gaze Equation’s thresholds $F_w \in \{1, 13/12, 6/5, 3/2\}$ classify the observation into UNOBSERVED, SUBLIMINAL, DETECTED, or LOCKED (§??). (e) *Trajectory*: for active systems (§??), the iteration dynamics produce a trajectory through the FQG cells, classified per-step by (a)–(d) applied at each step. The five-tuple is complete because every $X \in \Sigma$ is a $P \circ D \circ T$ configuration (Subsumption Law, §??), and the quadruple of inputs decomposes this configuration into its structural, geometric, methodological, and observational components. Completeness at sufficient resolution follows from the subsumption of inputs under the Four Paths (Theorem ??). □ □

Corollary 20.2 (Lattice Completeness Statement). *For every $X \in \Sigma$ there exists a complete lattice classification, all five components of which are forward-derivable from the three primitive Cardinals $\{P, D, T\}$ and the four manifold states $\{\{P, D, T\}, \{P, D\}, \{D, T\}, \{P, T\}\}$, with zero external axioms, zero tuning, and zero ad hoc additions.*

Remark 20.3 (Non-triviality of completion). The corollary is the non-trivial completion of the Sempaevum apparatus. Prior to the integration of the Four Paths, EML, Math-as-a-Domain, and the Complete Gaze Equation, the framework classified most but not all of Σ . The integration closes the four gaps — user-input method selection, essentially-infinite object handling

without limits, mathematics-as-a-domain including non-computables and undecidables, and observation/attention as a projectable functional — with the Complete Determination Theorem tying them together. The resulting classification applies uniformly to fictional or hypothetical configurations (Domain Validity Theorem, §??), experimentally observed phenomena, non-computable reals (Chaitin Ω at $d = 1$ octave via Path D.P), formally undecidable propositions (Gödel sentences via integrative-level classification), and active dynamical trajectories alike.

21 Falsifiable predictions

Any framework of structural significance must admit predictions that can, in principle, fail. The ET lattice admits at least the following.

21.1 The closure-versus-linear distinction in biochemistry

Any true biochemical closure cycle (one in which the product is identical to the starting substrate up to the catalytic turnover) has a step count n satisfying $n \in \{1, 2, 4, 8, 16, \dots\}$, that is, n is a power of 2. Any linear biochemical pathway (no closure) has step count n not a power of 2.

The prediction is testable against every metabolic pathway that has been completely characterised biochemically. A closure cycle with step count $n \in \{3, 5, 6, 7, 9, 10, 11, \dots\}$ would falsify the prediction as stated.

21.2 Orbital resonances on low- d attractors

Stable mean-motion resonances in gravitational systems preferentially occupy sublattice families $d \in \{1, 2, 3, 4, 6\}$; the $d = 12$ family hosts only transient or unstable resonances.

The prediction is testable against the ensemble of known mean-motion resonances in the Solar System and in extrasolar multi-planet systems. A stable long-lived resonance at $d = 12$ would require explanation.

21.3 Fine-structure constant lattice identity

The lattice coordinates of α^{-1} at $N = 27720$ are $(k, d) = (196768, 315)$ independent of the precise measured value of α^{-1} within the Parker–Morel experimental window.

The prediction is that measurements of α^{-1} whose central values differ by up to several parts per billion all project to the same integer lattice coordinates at the universal lattice $N = 27720$. The coordinates change only at a precision of order $|\varepsilon| \cdot N/1200 = 0.002 \cdot 27720/1200 \approx 0.046$ parts per billion in α^{-1} — below current experimental resolution.

21.4 New sublattice identification

The prediction of a *new* sublattice attractor is the strongest form of test. The lattice predicts, for instance, that the $d = 35 = 5 \cdot 7$ composite family at the universal lattice $N = 420$ corresponds to phenomena that simultaneously require five-fold and seven-fold symmetry. One realisation is the icosahedral viral capsid of triangulation number $T = 7$, which has $60 \cdot 7 = 420$ protein subunits [?]. A second test would be a prediction of the minimum viable genome size at approximately 420 genes; observed values for minimal genomes (*Mycoplasma genitalium* at ~ 470 , *JCVI-syn3.0* at ~ 473 [?]) lie slightly above this threshold, consistent with the prediction.

22 Conclusion

We have identified, characterised, and verified a single mathematical object — the Sempaevum — as the lossless rendering of the totality Σ on the multiplicative manifold. The identification rests on Theorem ?? (the terminal-object property under the Subsumption Law) and is supported by eight self-referential closure properties that no candidate non-trivial mathematical structure exhibits simultaneously.

The Sempaevum is forward-derived from three irreducible primitive categories (P, D, T) and a single master identity $3 = 3 = 3 = \Sigma$. At the resolution $N = 12$ — forced by three independent derivations (§??) — it admits three categorically distinct minimal generators: Webb’s n -valued stroke (1935) on the discrete-logical side; the palindromic cascade on the discrete-multiplicative side; the EML operator (Odrzywółek 2026) on the continuous-elementary side (Theorem ??). The projection formula $k = \text{round}(N \log_2 r)$ itself decomposes into PDT atomic operations (Theorem ??) whose continuous- D content is implementable entirely as finite EML trees.

Essentially-infinite and non-computable mathematical objects are handled on the Sempaevum without limit-approaches via Path D (Theorem ??). Chaitin’s Ω sits at a definite address $(-84, 1, +13.794\text{¢})$ — the $d = 1$ octave sublattice (Theorem ??). ZF set theory sits at $(+36, 1, 0\text{¢})$ — the unique zero-descriptor-gap formal-system address (Corollary ??). Gödel sentences admit an integrative-level-dependent classification on the Sempaevum (Theorem ??). Large cardinals are placed by consistency-strength (Theorem ??).

The self-projection identity (Theorem ??) lands the Sempaevum’s own four defining constants $\{N, 1/N, K, 1/K\}$ onto a single sublattice position ($d = 12, |\varepsilon| = 1.955\text{¢}$), the Koide attractor. The Locked threshold of the Complete Gaze Equation (Theorem ??) coincides with this position (Corollary ??): maximum observer binding crystallises the target onto the manifold’s own self-consistency point. The Losslessness Theorem (Theorem ??) establishes that the representation map $r \mapsto (k, d, \varepsilon)$ is a bijection at every finite resolution and pullback-exact in the LCM-tower limit $N \rightarrow \infty$.

The empirical confrontation (§??) presents the Sempaevum’s fingerprint in particle physics (Koide ratio to six parts per million; fine-structure constant to sub-parts-per-billion under a closed-form identity matching recent LatComp 2025 measurements), atomic and molecular structure, musical intonation (the Pythagorean comma as an intrinsic $d = 12$ quantity), geometry (bond angles, polygon symmetry, crystal systems), biochemistry (Krebs and glycolysis step counts), and celestial mechanics. The clustering of structural ratios on small rationals is not a chance coincidence but the natural consequence of the Sempaevum’s finite low- d attractor set.

Relation to existing mathematics is complementary in every direction: Fourier analysis classifies frequencies, the Sempaevum classifies frequency ratios by sublattice family; Riemannian geometry describes curvature, the Sempaevum relates curvature to classification via $C(n) = n^2(n^2 - 1)/12$ with the base variance $V = 1/12$ as the denominator; number theory supplies the Gaussian-prime decomposition that controls the sublattice structure on the complex axis; Gödel–Turing–Chaitin incompleteness phenomena acquire definite lattice addresses rather than remaining outside structural classification.

The construction uses no tunable parameters, no data-fitting, and no external axioms. The integer $N = 12$, the temperament step $2^{1/12}$, the six simple sublattice families $\{1, 2, 3, 4, 6, 12\}$, the doubling law $\tau(N_\ell) = 6 \cdot 2^\ell$, the Koide threshold $K = 2/3$, and the fine-structure value $A_0 = 137$ are all forced. The predictions of §?? are testable and falsifiable.

We close with the two identities from which every result descends:

For every exception there is an exception, except the exception.

$$P \circ D \circ T = E \qquad 3 = 3 = 3 = \Sigma$$

References

- [1] Y. Koide, *Fermion–boson two-body model of quarks and leptons and Cabibbo mixing*, Lett. Nuovo Cimento **34** (1982) 201–205.
- [2] C. J. Cohen and E. C. Hubbard, *Libration of the close approaches of Pluto to Neptune*, Astron. J. **70** (1965) 10–13.
- [3] B. Alberts *et al.*, *Molecular Biology of the Cell*, 6th ed., Garland Science, New York, 2014.
- [4] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*, W. H. Freeman, San Francisco, 1973, §13.
- [5] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, *CODATA recommended values of the fundamental physical constants: 2022*, Rev. Mod. Phys. **97** (2025) 025002.
- [6] *Lattice Compendium 2025 reference value for α^{-1}* , in: M. J. Muller, *Exception Theory Lattice Compendium*, manuscript, 2025.
- [7] D. L. D. Caspar and A. Klug, *Physical principles in the construction of regular viruses*, Cold Spring Harbor Symp. Quant. Biol. **27** (1962) 1–24.
- [8] C. A. Hutchison *et al.*, *Design and synthesis of a minimal bacterial genome*, Science **351** (2016) aad6253.
- [9] M. J. Muller, *The ET Universal Projection Guide — How to Project Anything onto the ET Lattice, Across Every Domain, Statically and Actively*, manuscript, version 2.2, 2026.
- [10] M. J. Muller, *Exception Theory: The Three Operational Tools*, manuscript, 2026.
- [11] D. L. Webb, *Generation of any n -valued logic by one binary operation*, Proc. Nat. Acad. Sci. USA **21** (1935) 252–254.
- [12] H. M. Sheffer, *A set of five independent postulates for Boolean algebras, with application to logical constants*, Trans. Amer. Math. Soc. **14** (1913) 481–488.
- [13] A. Odrzywólek, *All elementary functions from a single operator*, arXiv:2603.21852v2 [cs.SC], April 2026. Archival DOI: 10.5281/zenodo.19183008. Code: <https://github.com/VA00/SymbolicRegressionPackage>.
- [14] M. W. Reimann, M. Nolte, M. Scolamiero, K. Turner, R. Perin, G. Chindemi, P. Dłotko, R. Levi, K. Hess, H. Markram, *Cliques of Neurons Bound into Cavities Provide a Missing Link between Structure and Function*, Frontiers in Computational Neuroscience **11**:48 (2017). DOI: 10.3389/fncom.2017.00048.
- [15] T. Needham, *Visual Complex Analysis*, Oxford University Press, 1997, Chap. 3 (Möbius transformations) and Chap. 6 (the Riemann sphere and Lorentz group).